

Transport Processes and Cardiometabolic Dysfunction

The Functional Medicine Matrix



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APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Sanjay Bhojraj, MD, disclosed no relevant financial relationships with ineligible companies.

Regina Druz, MD, MBA, was a speaker for Boston Heart Diagnostics.

Robert Luby, MD, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

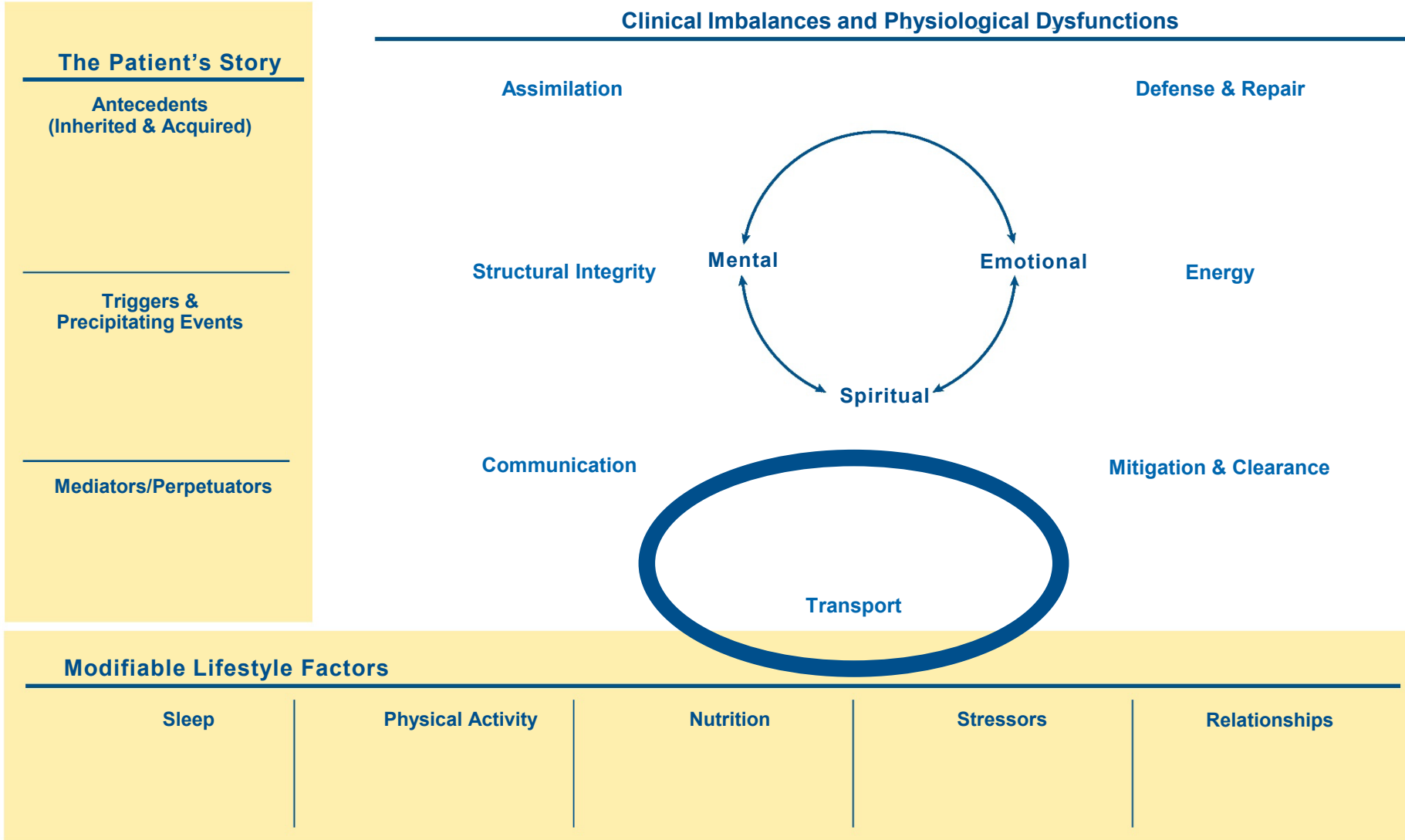
1. Explain how the clinical findings of cardiometabolic syndrome are due to the effects of excess insulin and the variable thresholds of different tissues to insulin resistance.
2. Explain how insulin resistance is the consequence of dysfunction of intracellular transport processes.
3. Explain the relationships among stress, inflammation, obesity, excess cortisol, endotoxemia (due to GI dysfunction), and insulin resistance.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Identify and document the patient's clinical findings, antecedents, triggers, and mediators of insulin resistance, dysglycemia, and cardiometabolic syndrome on the IFM Timeline and Matrix.
2. Explain to a patient how their diet, lifestyle behaviors, chronic stress, and other factors in their life (antecedents, triggers, and mediators) represent root causes of the insulin resistance that underpin their clinical findings and current concerns.
3. Co-prioritize and initiate dietary, lifestyle, and gut health interventions as foundational therapeutic interventions to address the patient's root cause contributors (ATMs) of insulin resistance.
4. Initiate foundational supplements to address insulin resistance and cardiometabolic syndrome as clinically indicated.

Functional Medicine Matrix



Clinical Focus

- Overview of the functional medicine perspective of dysinsulinemia and cardiometabolic syndrome.



Cardiovascular vs Cardiometabolic

Cholesterol-Rich
Disorders



VASCULAR
Focus



Triglyceride-Rich
Insulin-Resistant
Disorders



METABOLIC
Focus



Comprehensive Risk Reduction

Long Description: Cardiovascular vs Cardiometabolic

- Cardiovascular disease is distinguished from cardiometabolic disease in that the former are cholesterol-rich disorders with a vascular focus and the latter are triglyceride-rich disorders of insulin resistance with a metabolic focus.

Triglyceride-Rich
Insulin-Resistant
Disorders

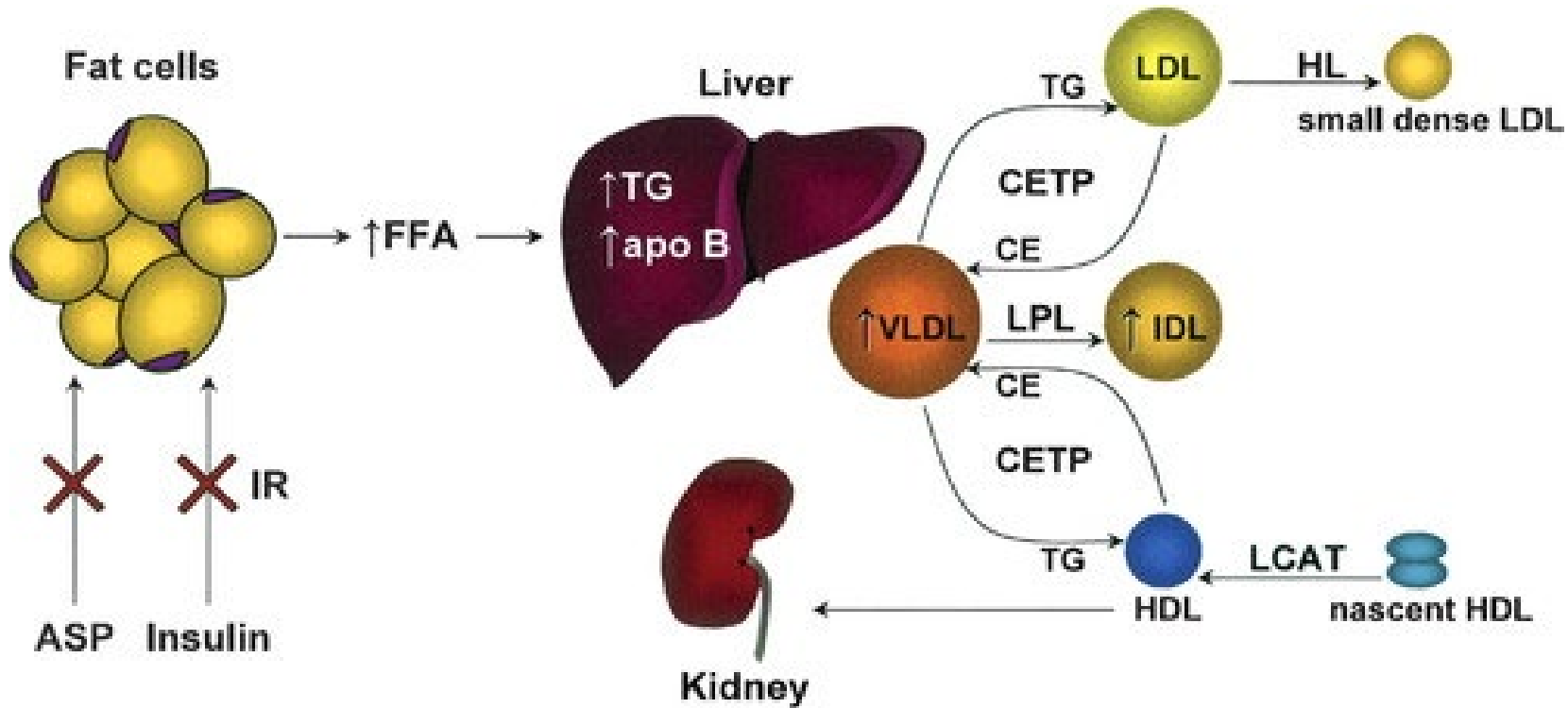


Metabolic Focus
Dyslipidemic Triad



Comprehensive Risk Reduction

Dyslipidemic Triad



Long Description: Dyslipidemic Triad

Insulin resistance drives hepatic uptake of free fatty acids, leading to increased triglyceride and VLDL production. Cholesterol ester transport protein moves triglycerides from VLDL to LDL and HDL, resulting in small, dense LDL particles and reduced HDL levels due to increased clearance through the kidneys.

Bodily tissues vary greatly in their thresholds of insulin resistance.

The tissues that are primarily involved in the dyslipidemic triad: skeletal muscle, liver, and adipose tissue.

These three tissues express the lowest threshold for insulin resistance.

Most other tissues remain insulin sensitive when skeletal muscle, liver, and adipose tissue have become insulin resistant.

Therefore, many of the adverse effects of dysinsulinemia are a result of the action of **elevated insulin levels on tissues that remain sensitive to insulin.**

Insulin's Role in Muscle Tissue

Increases glucose uptake

- Insulin stimulates the translocation of glucose transporter type 4 (GLUT4) to the cell membrane, facilitating the entry of glucose into muscle cells.

Enhances glycolysis

- Insulin increases the activity of key glycolytic enzymes, such as hexokinase and 6-phosphofructokinase, promoting the conversion of glucose to pyruvate.

Promotes glycogen synthesis

- Insulin activates glycogen synthase, the enzyme responsible for converting glucose to glycogen, storing energy in the muscle tissue.

Inhibits glycogen breakdown

- Insulin suppresses the activity of glycogen phosphorylase, the enzyme that breaks down glycogen, thus preventing the release of glucose from glycogen stores.

Decreases fatty acid oxidation

- Insulin inhibits the breakdown of fats (lipolysis) and promotes the storage of fatty acids as triglycerides, thereby reducing fatty acid oxidation in muscle tissue.

Increases protein synthesis

- Insulin stimulates the uptake of amino acids and enhances the translation of mRNA, leading to the increased production of proteins in muscle cells.

Insulin's Impact on Adipose Tissue

Stimulates glucose uptake and glycolysis

- Insulin enhances the utilization of glucose in adipose tissue, promoting its uptake and conversion to energy through glycolysis.

Inhibits lipolysis

- Insulin suppresses the breakdown of stored triglycerides (lipolysis) in adipose tissue, thereby reducing the release of free fatty acids into the bloodstream.

Promotes fatty acid and triglyceride synthesis

- Insulin stimulates the synthesis of fatty acids and triglycerides in adipose tissue, facilitating the storage of excess energy as fat.

Facilitates the uptake of triglycerides from the blood

- Insulin enhances the ability of adipose tissue to take up and store triglycerides circulating in the bloodstream.

Insulin's Actions in the Liver

Suppresses endogenous glucose production

- Insulin inhibits gluconeogenesis and glycogenolysis, reducing the liver's output of glucose.

Stimulates glycogen synthesis

- Insulin promotes the storage of glucose in the liver as glycogen.

Promotes lipid synthesis

- Insulin enhances the conversion of glucose and other substrates into fatty acids and triglycerides.

Inhibits fatty acid oxidation

- Insulin suppresses the breakdown of fatty acids for energy, favoring their storage.

Insulin's Vascular Effects

Insulin-induced vasodilation

- Insulin stimulates the release of nitric oxide, leading to relaxation of vascular smooth muscle cells and increased blood flow.

Enhanced glucose disposal in muscle

- Insulin-mediated vasodilation increases the delivery of glucose to skeletal muscle, facilitating its uptake and utilization.

Sodium-potassium pump activation

- Insulin triggers the activation of the sodium-potassium pump, which contributes to the vasodilatory effect by regulating ion gradients and membrane potential.

Intracellular calcium signaling

- Insulin stimulates the release and sequestration of intracellular calcium, leading to the activation of endothelial nitric oxide synthase and vasodilation.

Insulin Modulates Gene Transcription and Expression That Regulate the Following Processes:

- Glucose metabolism
- Fat mobilization and synthesis
- Protein metabolism
- Mitochondrial function
- Cytokine synthesis
- Oxidative stress and inflammatory response
- Adrenal and sex hormone metabolism

Elevated levels of insulin may cause dysregulation of many other bodily tissues beyond glucose regulation.

References

INSULIN-SENSITIVE TISSUE

Adipose Tissue/Muscle:

- Dimitriadis G, Mitrou P, Lambadiari V, Maratou E, Raptis SA. Insulin effects in muscle and adipose tissue. *Diabetes Res Clin Pract.* 2011;93(Suppl 1):S52-S59. doi:10.1016/S0168-8227(11)70014-6

Adrenal Glands:

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INSULIN-SENSITIVE TISSUE

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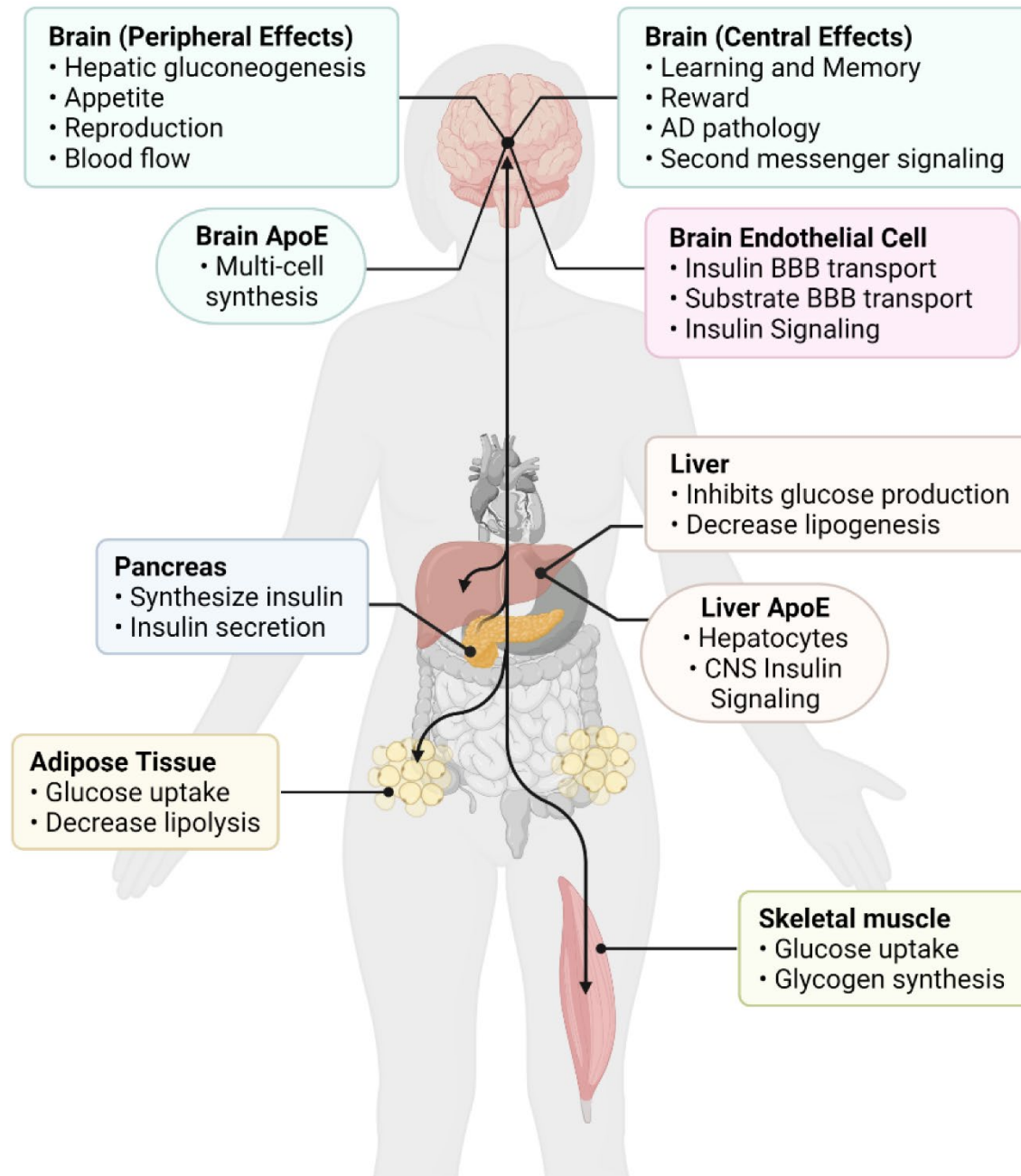
Ovaries:

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Testes:

- Leisegang K, Henkel R, Agarwal A. Obesity and metabolic syndrome associated with systemic inflammation and the impact on the male reproductive system. *Am J Reprod Immunol*. 2019;82(5):e13178. doi:10.1111/aji.13178

Insulin Resistance: Effects on Peripheral Tissues and the Brain



Long Description: Insulin Resistance: Effects on Peripheral Tissues and the Brain

- Brain (peripheral effects)
 - Hepatic gluconeogenesis
 - Appetite
 - Reproduction
 - Blood Flow
- Brain (Central effects)
 - Learning and memory
 - Reward
 - A D pathology
 - Second messenger signaling
- Brain Apo E
 - Multi-cell synthesis
- Brain Endothelial Cell
 - Insulin blood brain barrier transport
 - Substrate blood brain barrier transport
 - Insulin signaling
- Liver
 - Inhibits glucose production
 - Decrease lipogenesis
- Liver Apo E
 - Hepatocytes
 - C N S insulin signaling
- Pancreas
 - Synthesize insulin
 - Insulin secretion
- Adipose tissue
 - Glucose uptake
 - Decrease lipolysis
- Skeletal muscle
 - Glucose uptake
 - Glycogen synthesis

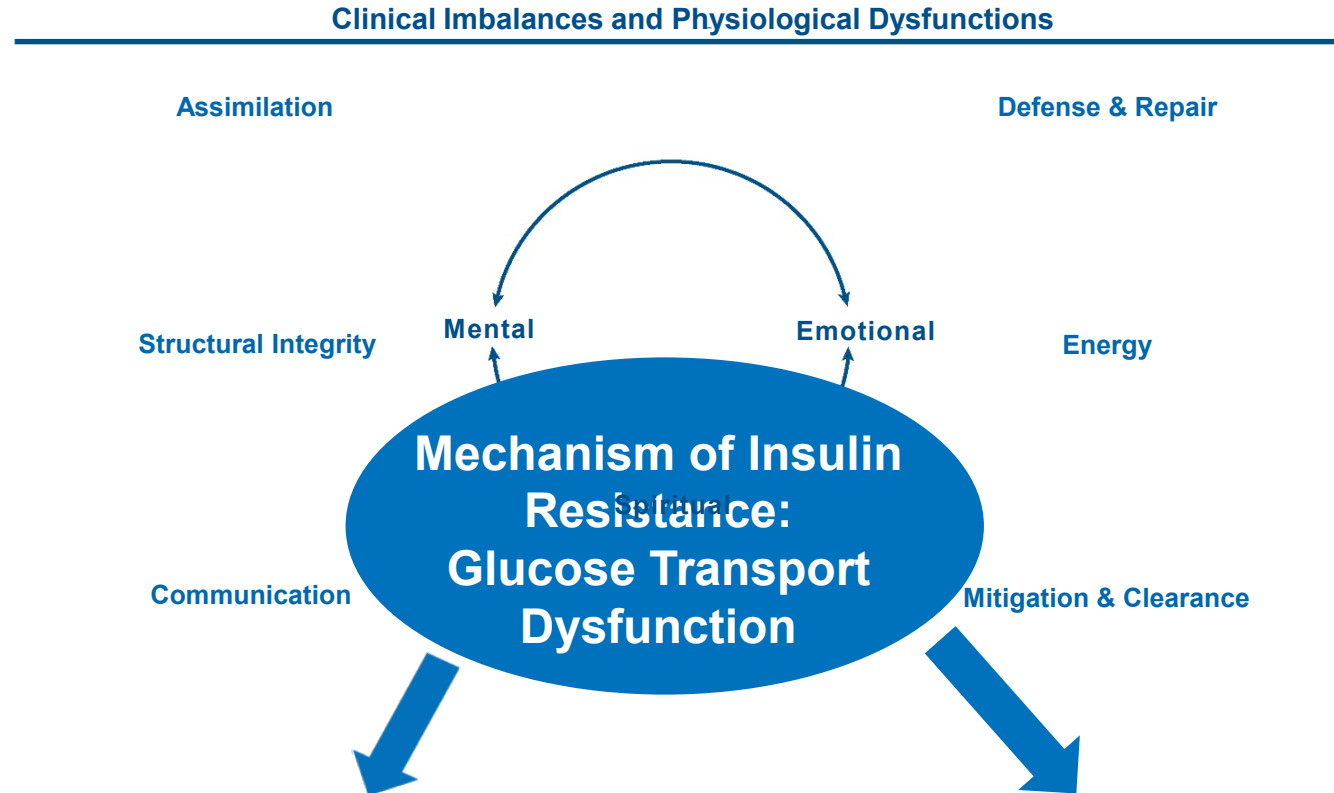
Functional Medicine Matrix

The Patient's Story

Antecedents
(Inherited & Acquired)

Triggers &
Precipitating Events

Mediators/Perpetuators

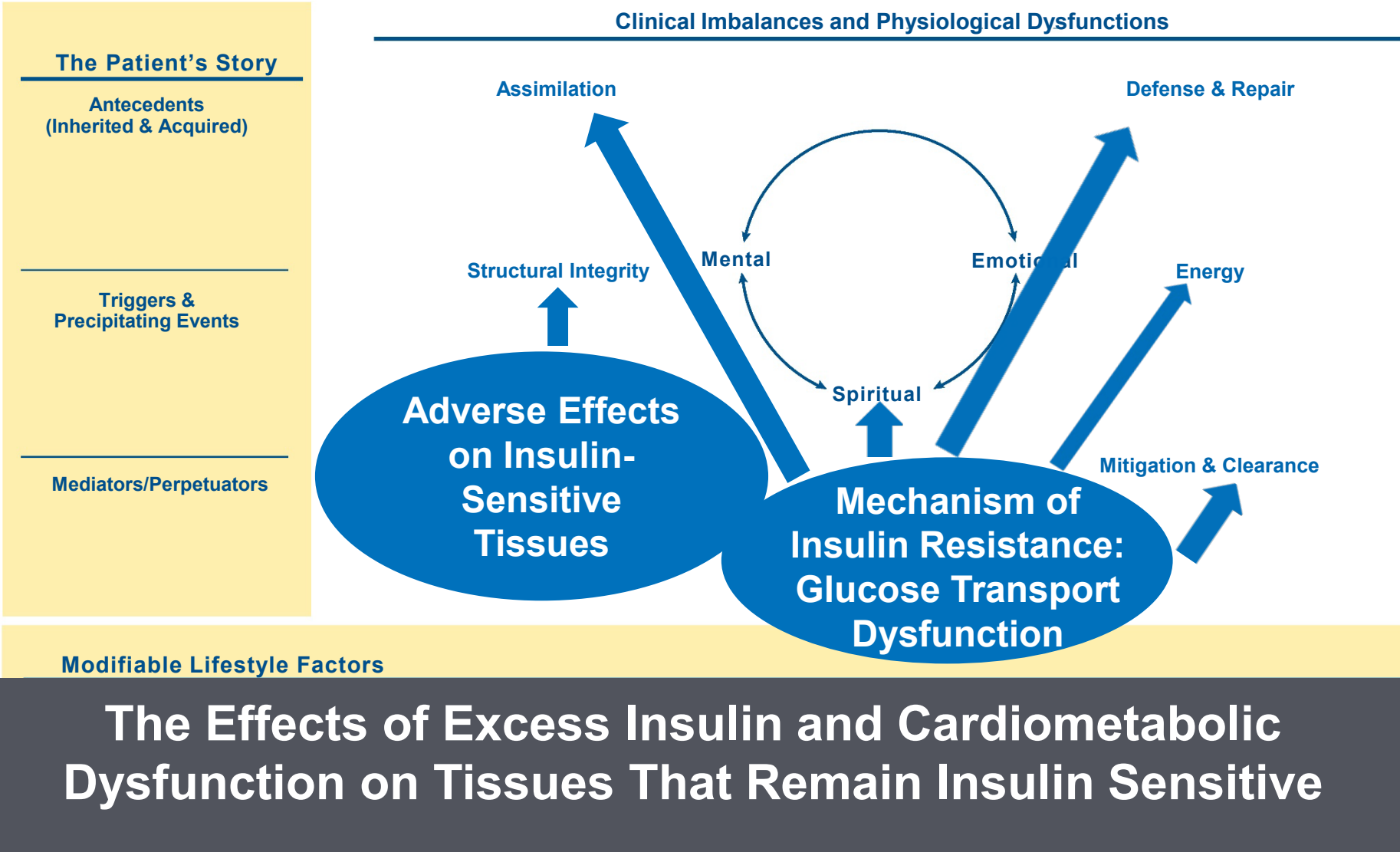


As insulin levels increase, some tissues become insulin resistant while others remain insulin sensitive.

Long Description: **Mechanism of Insulin Resistance: Glucose Transport Dysfunction**

Glucose transport dysfunction is the mechanism of insulin resistance, which lives in the transport node of the Functional Medicine Matrix.

Functional Medicine Matrix



Long Description: **The Effects of Excess Insulin and Cardiometabolic Dysfunction on Tissues That Remain Insulin Sensitive**

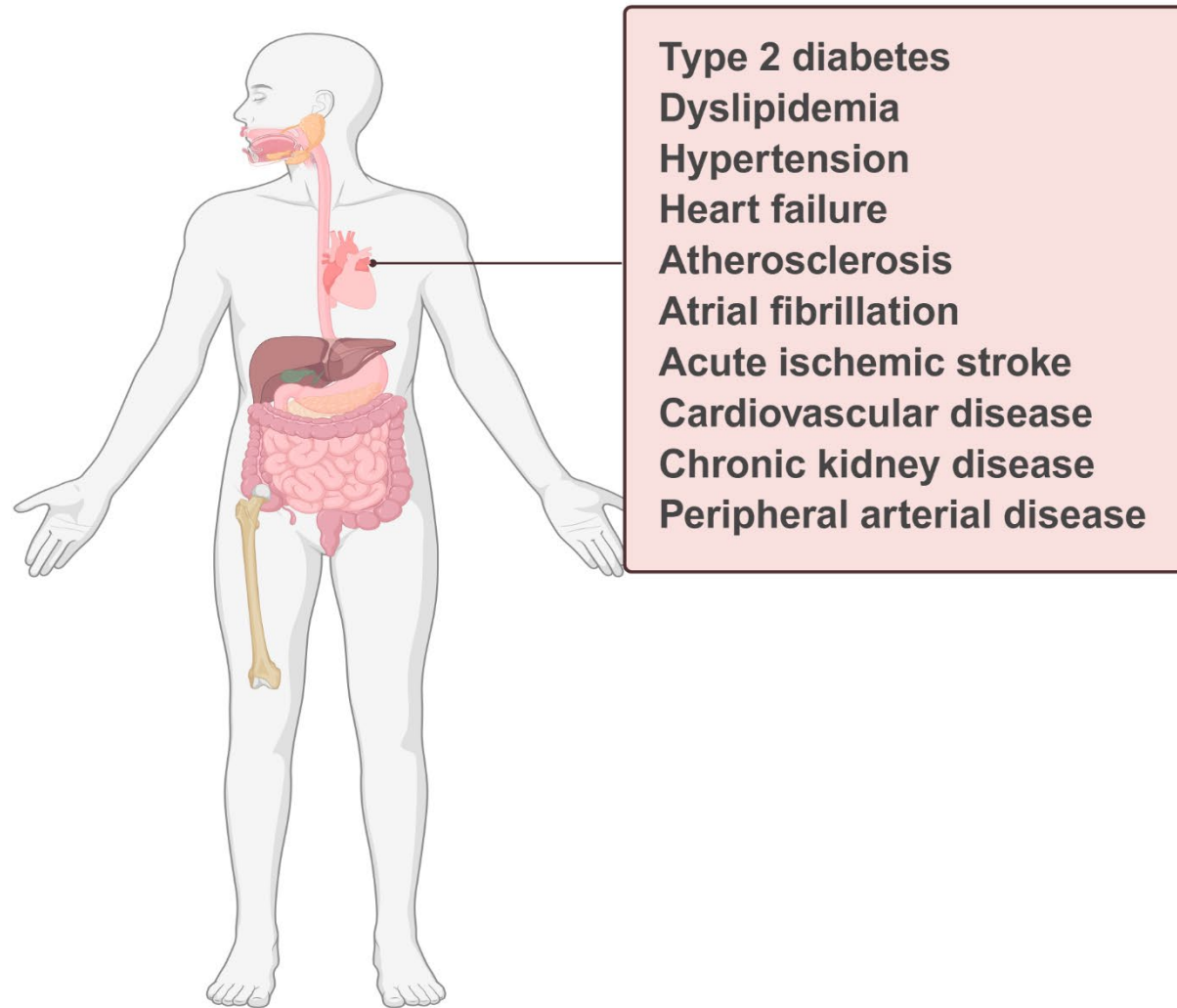
Insulin resistance impacts the assimilation, defense and repair, energy, mitigation and clearance, mental, emotional, and spiritual nodes of the matrix. Adverse effects on insulin-sensitive tissues lives in the communication node and impacts the structural integrity node.

Clinical Focus

- Clinical findings of dysinsulinemia and cardiometabolic syndrome.



The “Cardio” in Cardiometabolic Syndrome: Chronic Conditions Linked to the Syndrome

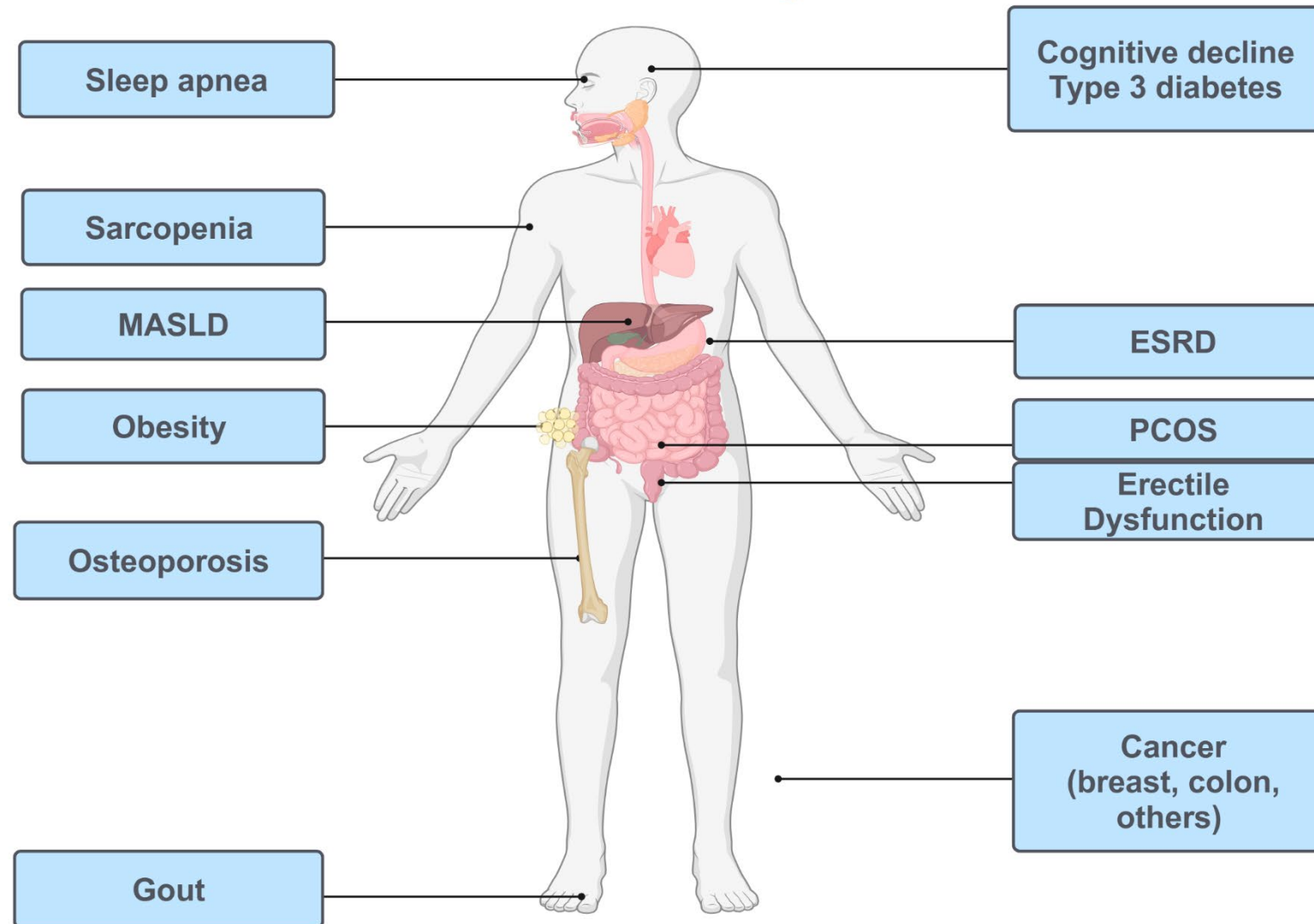


Long Description: The "cardio" in cardiometabolic syndrome: Chronic conditions linked to the syndrome

Chronic conditions linked to cardiometabolic syndrome:

- Type 2 diabetes
- Dyslipidemia
- Hypertension
- Heart failure
- Atherosclerosis
- Atrial fibrillation
- Acute ischemic stroke
- Cardiovascular disease
- Chronic kidney disease
- Peripheral arterial disease

Beyond the “Cardio” in Cardiometabolic Syndrome: Chronic Conditions Linked to the Syndrome



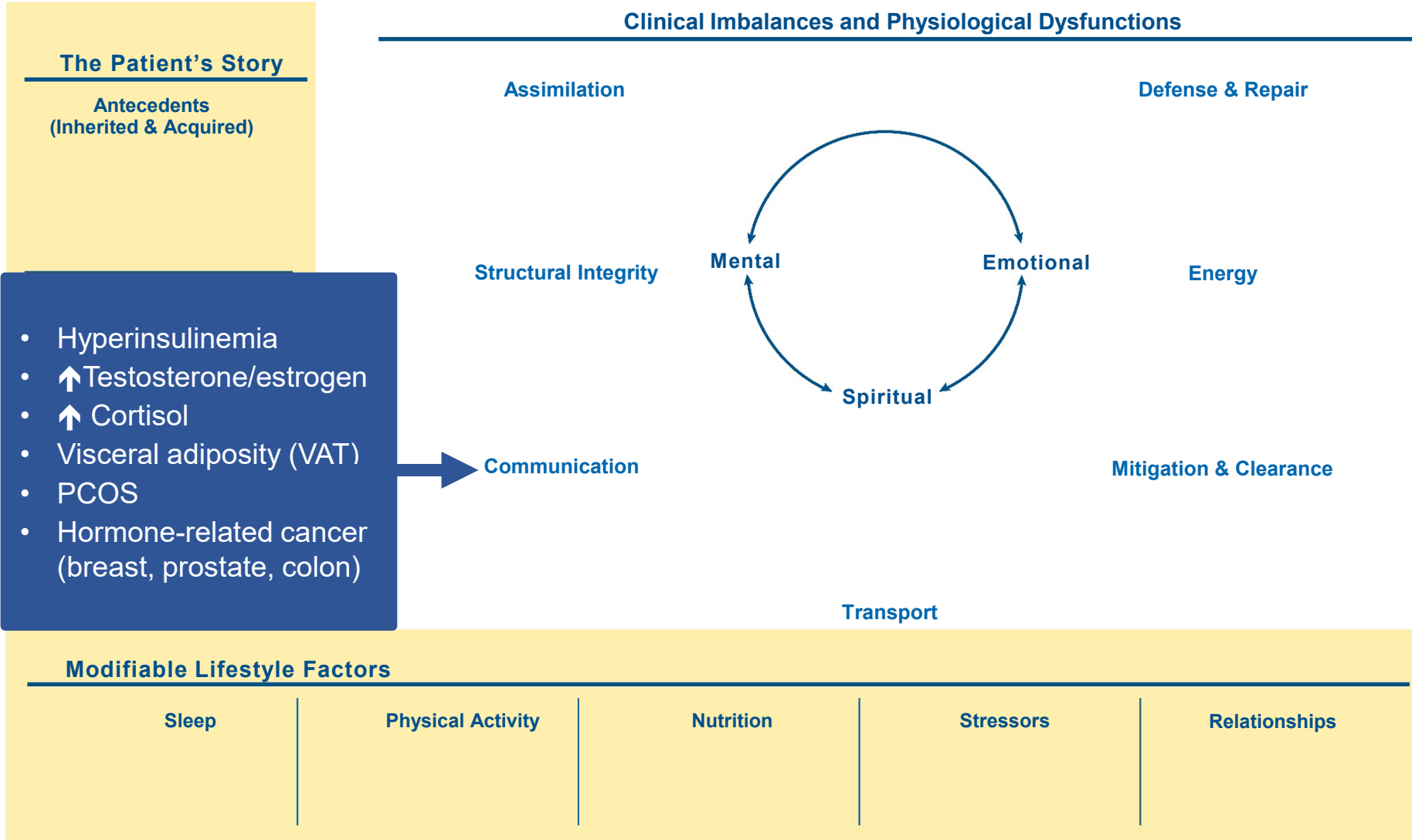
Long Description: The "cardio" in cardiometabolic syndrome: Chronic conditions linked to the syndrome

Chronic conditions linked to cardiometabolic syndrome beyond the cardiovascular system:

- Sleep apnea
- Cognitive decline in Type 3 diabetes
- Sarcopenia
- Metabolic dysfunction associated steatotic liver disease
- End-stage renal disease
- Obesity
- P C O S
- Erectile dysfunction
- Osteoporosis
- Gout
- Cancer

Functional Medicine Matrix

Clinical Findings of Dysinsulinemia Around the Matrix



See: "References: Clinical Findings of Cardiometabolic Syndrome"

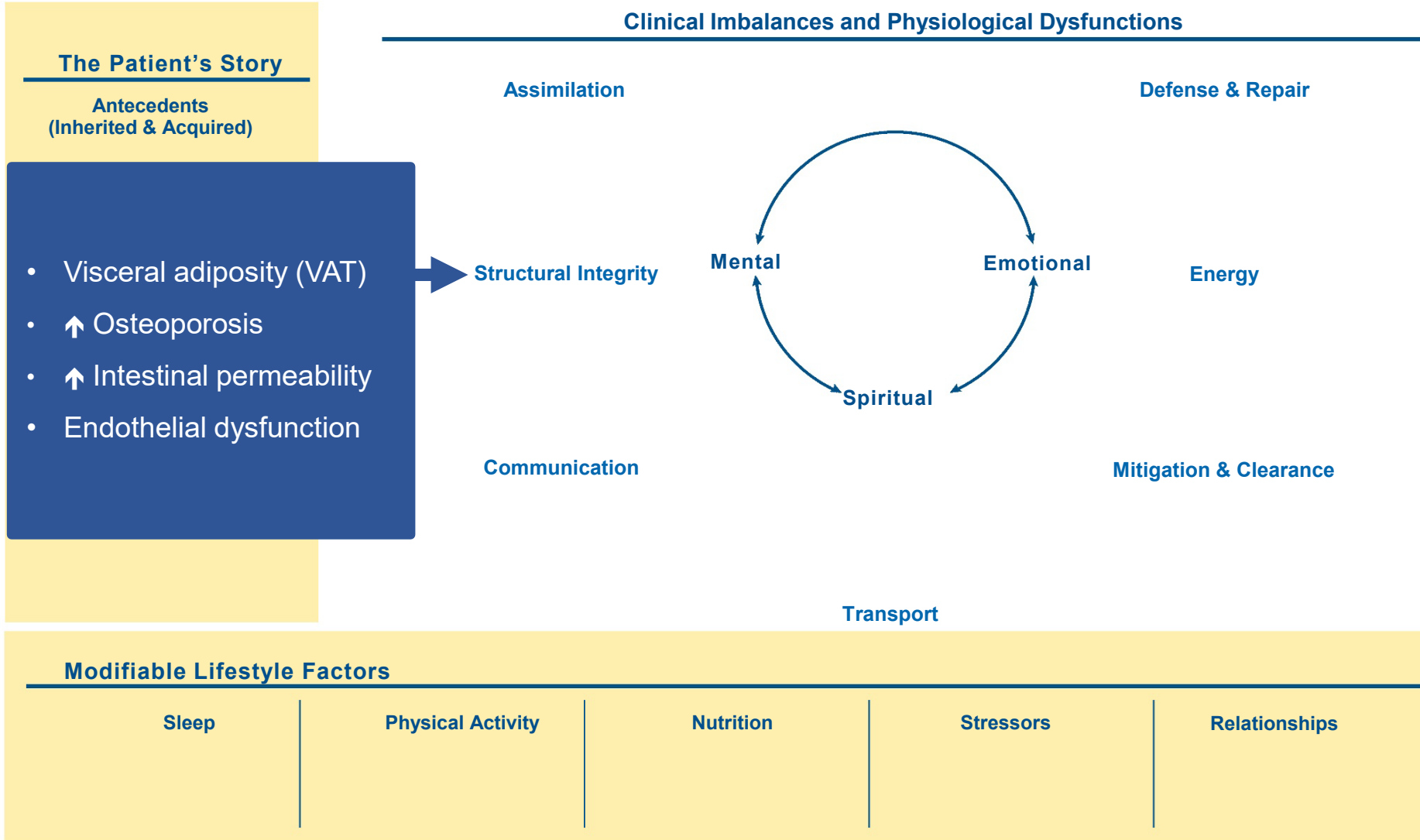
Long Description: Clinical Findings of Dysinsulinemia Around the Matrix

Communication node

- Hyperinsulinemia
- Increased testosterone and estrogen
- Increased cortisol
- Visceral adiposity
- P C O S
- Hormone-related cancers: breast, prostate, colon

Functional Medicine Matrix

Clinical Findings of Dysinsulinemia Around the Matrix



See: "References: Clinical Findings of Cardiometabolic Syndrome"

Long Description: Clinical Findings of Dysinsulinemia Around the Matrix

Structural integrity node

- Visceral adiposity
- Increased osteoporosis
- Increased intestinal permeability
- Endothelial dysfunction

Functional Medicine Matrix

- Adverse food reactions
- Endotoxemia
- Dysbiosis
- ↑ Intestinal permeability

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Transport

Triggers &
Precipitating Events

Mediators/Perpetuators

Modifiable Lifestyle Factors

Sleep

Physical Activity

Nutrition

Stressors

Relationships

Clinical Findings of Dysinsulinemia Around the Matrix

See: "References: Clinical Findings of
Cardiometabolic Syndrome"

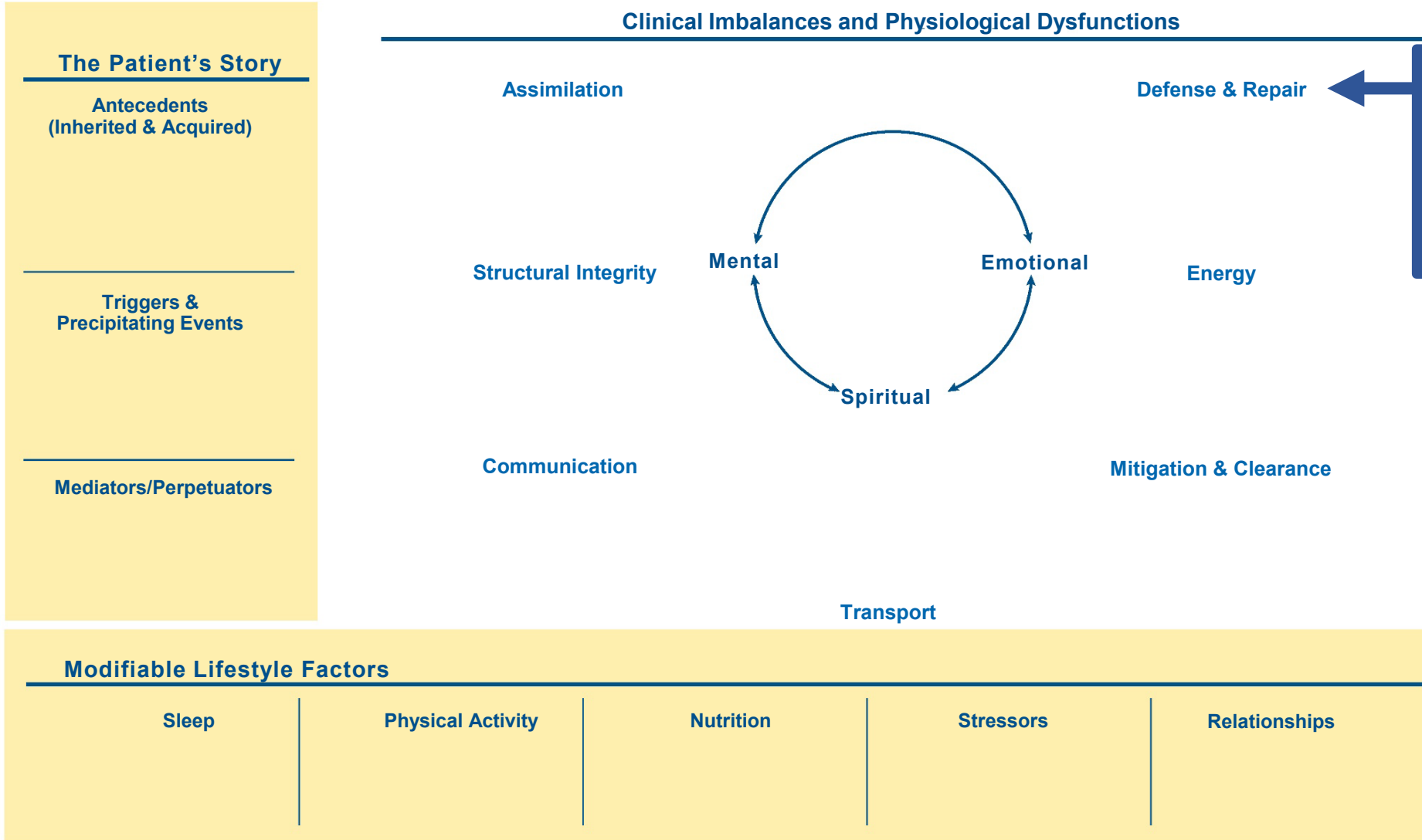
Long Description: Clinical Findings of Dysinsulinemia Around the Matrix

Assimilation node

- Adverse food reactions
- Endotoxemia
- Dysbiosis
- Increased intestinal permeability

Functional Medicine Matrix

Clinical Findings of Dysinsulinemia Around the Matrix



- Chronic infections
- Immune dysfunction
- Systemic inflammation

See: "References: Clinical Findings of Cardiometabolic Syndrome"

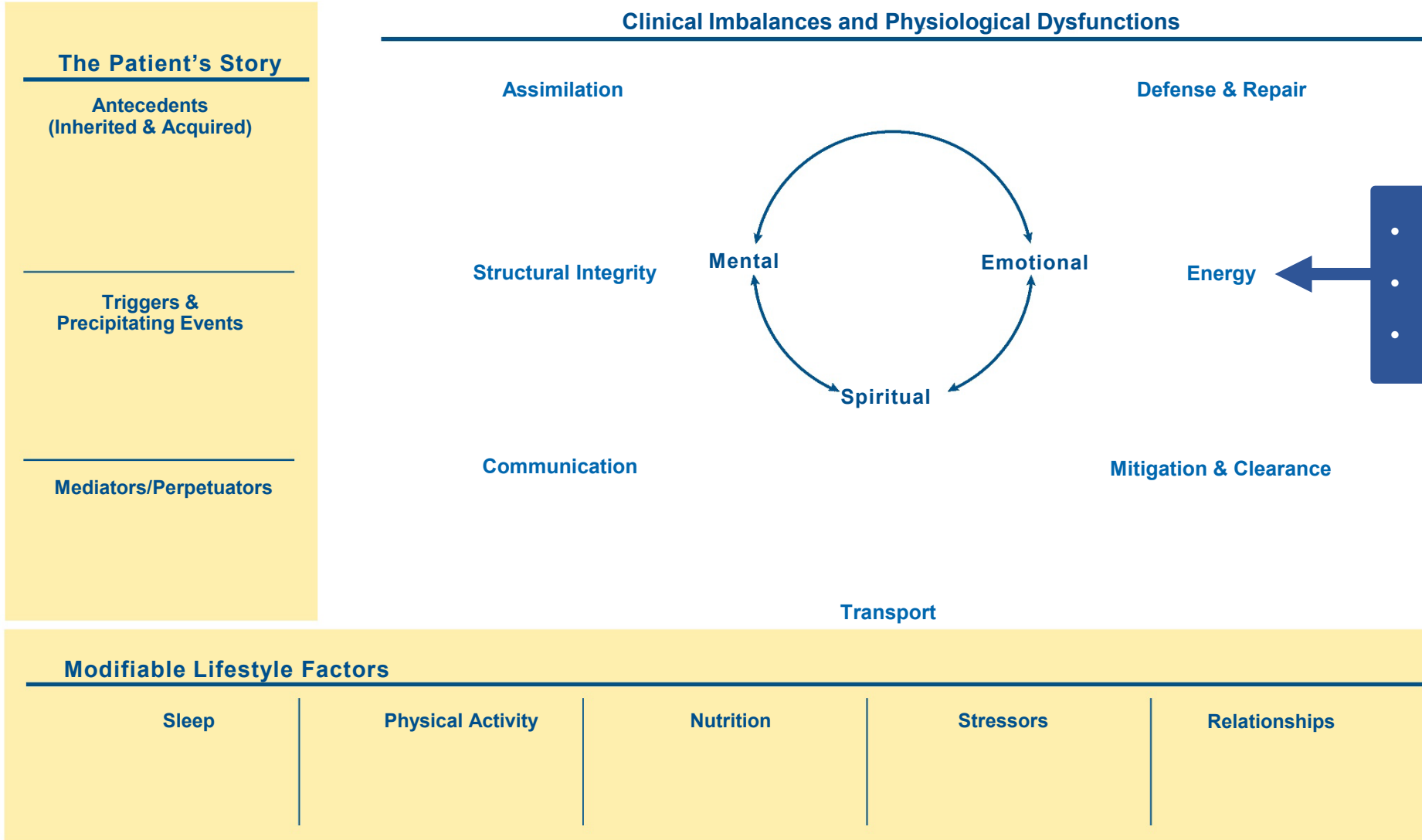
Long Description: Clinical Findings of Dysinsulinemia Around the Matrix

Defense and repair node

- Chronic infections
- Immune dysfunction
- Systemic inflammation

Functional Medicine Matrix

Clinical Findings of Dysinsulinemia Around the Matrix



See: "References: Clinical Findings of Cardiometabolic Syndrome"

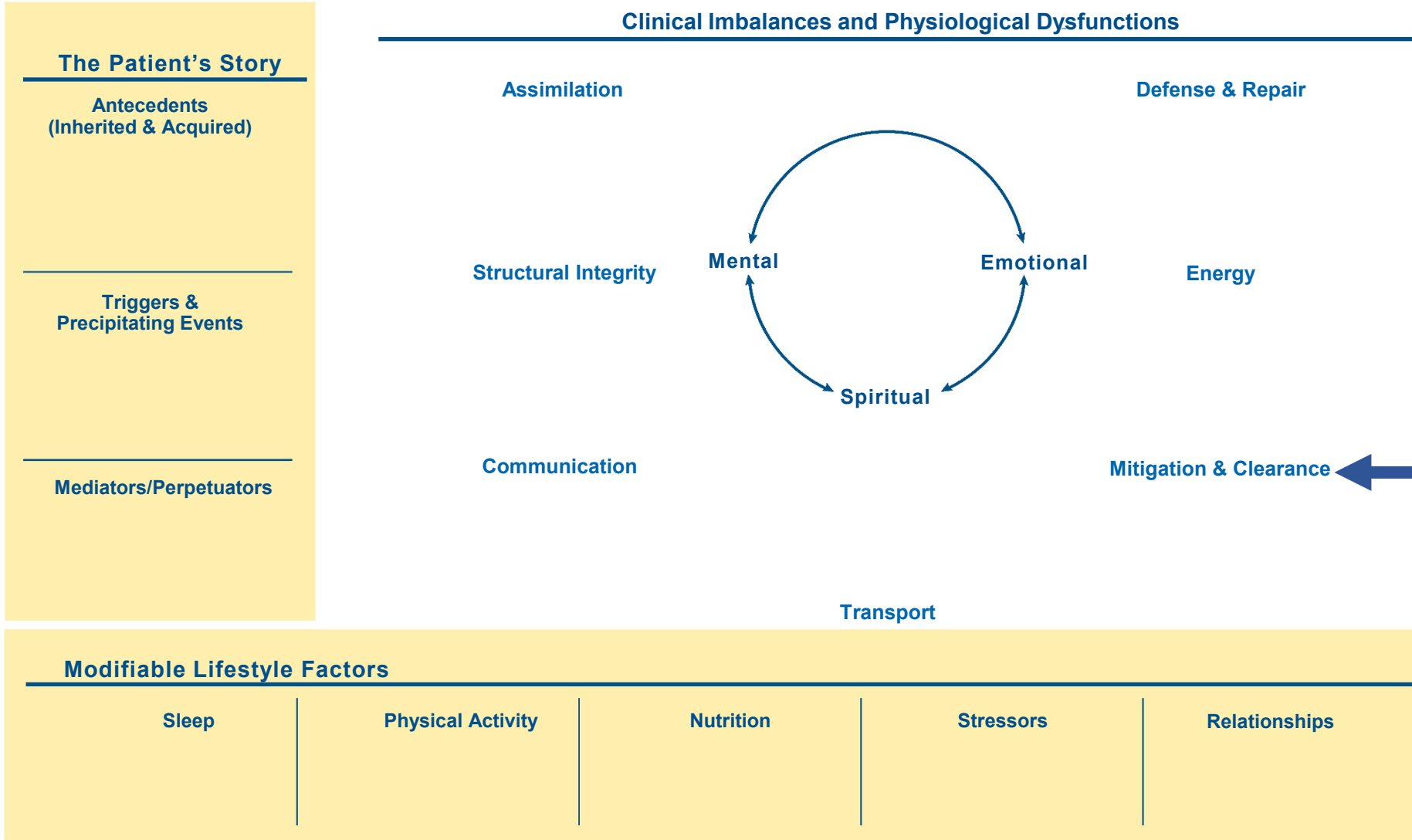
Long Description: Clinical Findings of Dysinsulinemia Around the Matrix

Energy node

- Increased oxidative stress
- Increased mitochondrial dysfunction
- Type 3 diabetes

Functional Medicine Matrix

Clinical Findings of Dysinsulinemia Around the Matrix



- Constipation
- Fatty liver/MASLD

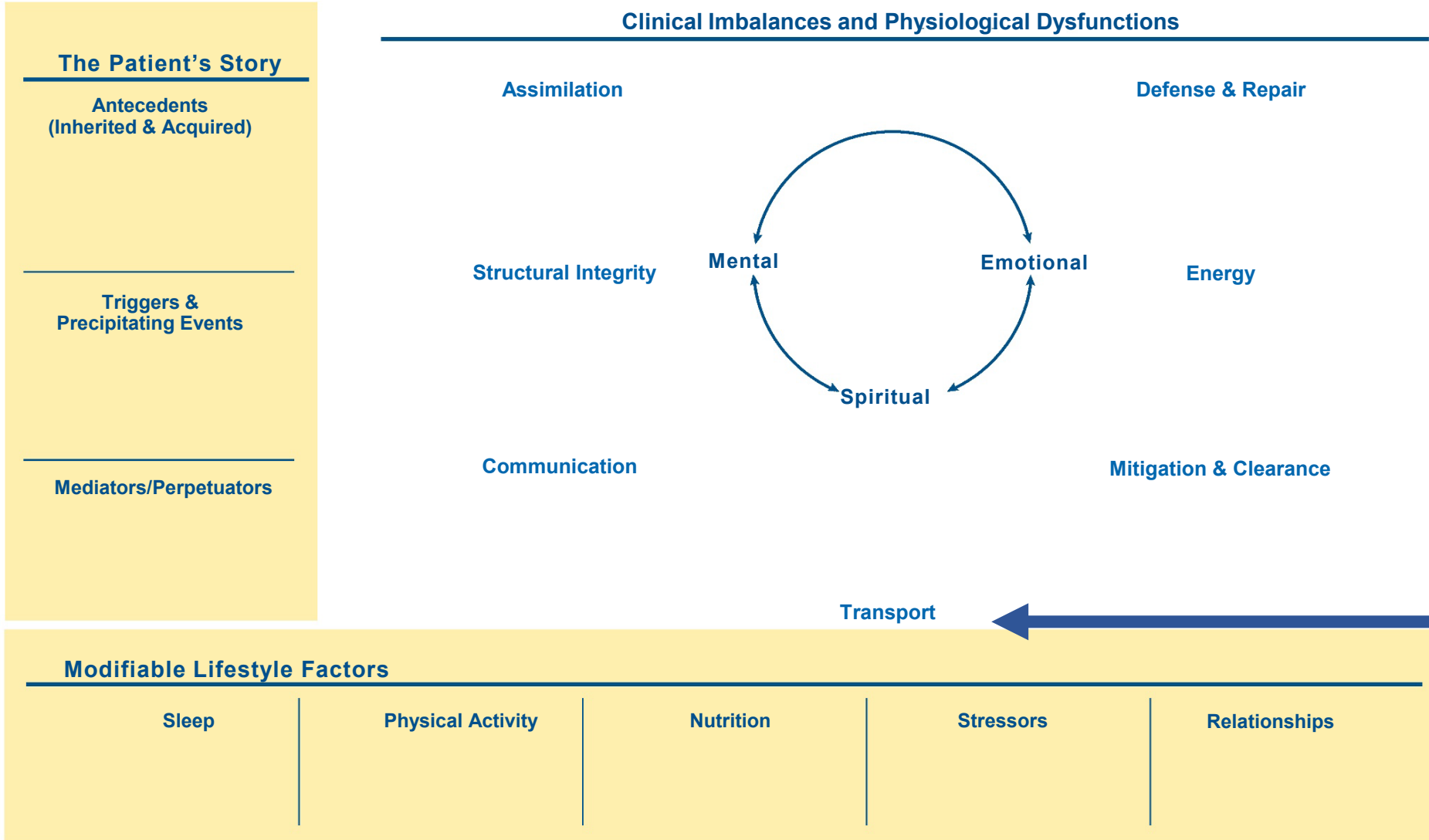
See: "References: Clinical Findings of Cardiometabolic Syndrome"

Long Description: Clinical Findings of Dysinsulinemia Around the Matrix

Mitigation and clearance node

- Constipation
- Fatty liver/ M A S L D

Functional Medicine Matrix



Clinical Findings of Dysinsulinemia Around the Matrix

- Type 2 diabetes
- Hypertension
- Atherosclerosis
- Cardiovascular disease
- Heart failure
- Atrial fibrillation
- Peripheral artery disease

See: "References: Clinical Findings of Cardiometabolic Syndrome"

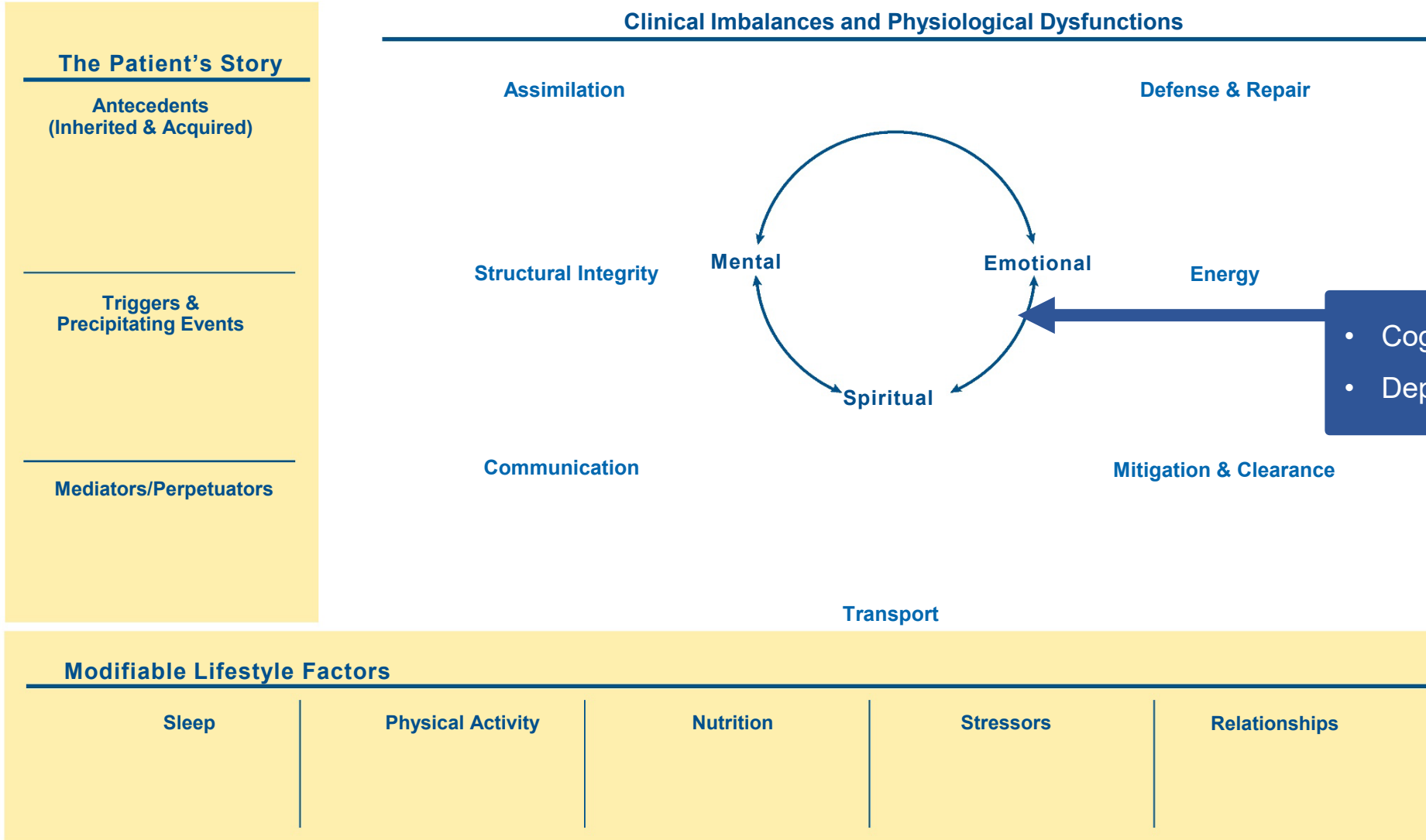
Long Description: Clinical Findings of Dysinsulinemia Around the Matrix

Transport node

- Type 2 diabetes
- Hypertension
- Atherosclerosis
- Cardiovascular disease
- Heart failure
- Atrial fibrillation
- Peripheral artery disease

Functional Medicine Matrix

Clinical Findings of Dysinsulinemia Around the Matrix



See: "References: Clinical Findings of Cardiometabolic Syndrome"

Long Description: Clinical Findings of Dysinsulinemia Around the Matrix

Mental, emotional, spiritual node

- Cognitive impairment
- Depression

References

CLINICAL FINDINGS OF CARDIOMETABOLIC SYNDROME

Assimilation:

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CLINICAL FINDINGS OF CARDIOMETABOLIC SYNDROME

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CLINICAL FINDINGS OF CARDIOMETABOLIC SYNDROME

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CLINICAL FINDINGS OF CARDIOMETABOLIC SYNDROME

Transport:

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Additional Clinical Findings

Dysinsulinemia:

- Acrochordon (skin tags)
- Acanthosis nigricans
- Decrease in pinprick, temperature, vibration, and pressure sensation (using a 10 g monofilament pressure sensation at the distal halluces)
- Skin ulceration
- Tinea pedis

Additional Clinical Findings

Cardio-Kidney-Metabolic (CKM) Syndrome:

The American Heart Association defines (CKM) syndrome as:
“a health disorder attributable to connections among obesity, diabetes, chronic kidney disease (CKD), and cardiovascular disease (CVD), including heart failure, atrial fibrillation, coronary heart disease, stroke, and peripheral artery disease.”

Adverse social determinants of health (SDOH) create an excessive burden on those with CKM syndrome. Consistent SDOH screening is recommended as well as addressing SDOH as part of the clinical care model.

Cardio-Kidney-Metabolic (CKM) Syndrome

The American Heart Association

CKM syndrome encompasses a broader spectrum of cardiovascular, kidney, and metabolic conditions, classified into stages.

Stage 0	Stage 1	Stage 2	Stage 3	Stage 4
• No CKM risk factors.	• Early clinical disease with mild symptoms.	• Metabolic risk factors or moderate-to high-risk chronic kidney disease.	• Subclinical CVD in CKM syndrome or risk equivalents.	• Clinical CVD in CKM syndrome.

References

CARDIO-KIDNEY-METABOLIC (CKM) SYNDROME

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- Kittelson KS, Junior AG, Fillmore N, da Silva Gomes R. Cardiovascular-kidney-metabolic syndrome – an integrative review. *Prog Cardiovasc Dis*. 2024;87:26-36. doi:10.1016/j.pcad.2024.10.012
- Ndumele CE, Neeland IJ, Tuttle KR, et al. A synopsis of the evidence for the science and clinical management of cardiovascular-kidney-metabolic (CKM) syndrome: a scientific statement from the American Heart Association. *Circulation*. 2023;148(20):1636-1664. doi:10.1161/CIR.0000000000001186
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Transport Processes and Cardiometabolic Dysfunction: Part 2

The Functional Medicine Matrix

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Anthropometrics and laboratory evaluation of metabolic syndrome.



Metabolic Syndrome Diagnostic Criteria

International Diabetes Federation

Must have:
Central obesity (considering ethnicity-specific waist circumference)

PLUS 2 out of 4 of the following factors:

Elevated Triglycerides

- >150 mg/dL
- Drug treatment of hypertriglyceridemia is an alternate criterion

Low HDL-C

- <40 mg in males
- <50 mg/dL in females

Elevated Blood Pressure

- Systolic BP ≥ 130 mm Hg and/or diastolic BP ≥ 85 mm Hg
- Drug treatment of hypertension is an alternate criterion

Elevated Fasting Glucose

- ≥ 100 mg/dL
- Drug treatment of elevated glucose is an alternate criterion

Waist Circumference by Ethnicity

ATP III and International Diabetes Federation

Ethnicity	Circumference – Male	Circumference – Female
USA (ATP III)	40 inches >102 cm	35 inches >88 cm
European SS African Mediterranean Middle Eastern	37 inches >92 cm	31 inches >80 cm
Asian Hispanic (IDF)	35 inches >90 cm	31 inches >80 cm



- Alberti KG, Zimmet P, Shaw J. Metabolic syndrome—a new world-wide definition. A consensus statement from the International Diabetes Federation. *Diabet Med*. 2006;23(5):469-480. doi:10.1111/j.1464-5491.2006.01858.x
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Impaired Glucose Tolerance/Insulin Resistance Laboratory Criteria

- Fasting blood glucose 100-125 mg/dl
- 2-hour post-prandial blood glucose 140-199 mg/dl
- HbA1c of 5.7-6.4 %

All signify insulin resistance and hyperinsulinemia with faltering reserve capacity (aka pre-diabetes).

A Simple Test for Insulin Resistance

TG/HDL
Ratio

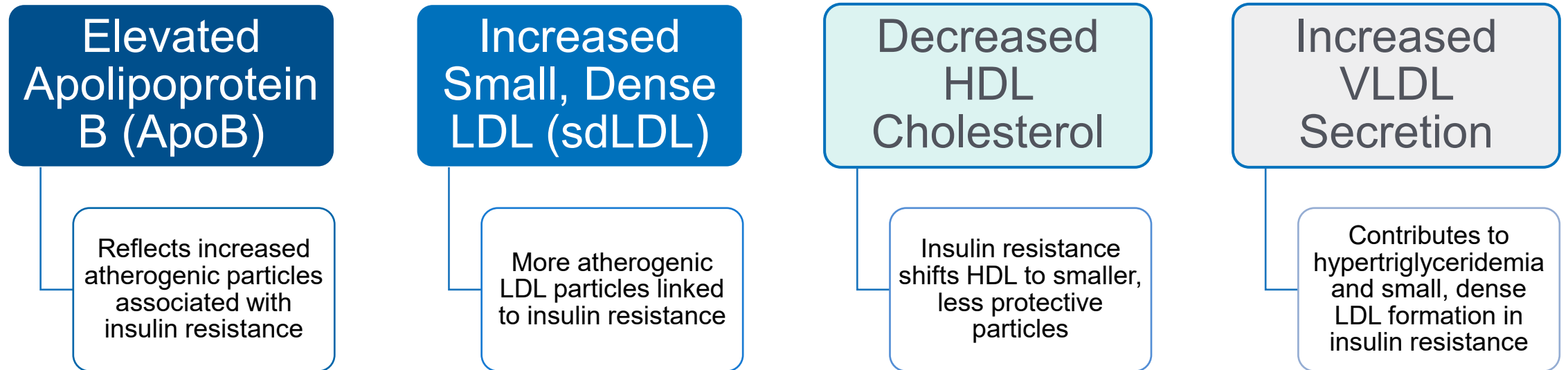
- <3.0 mg/dL is normal
- ≥ 3.0 mg/dL is suggestive of insulin resistance

References

TG/HDL RATIO

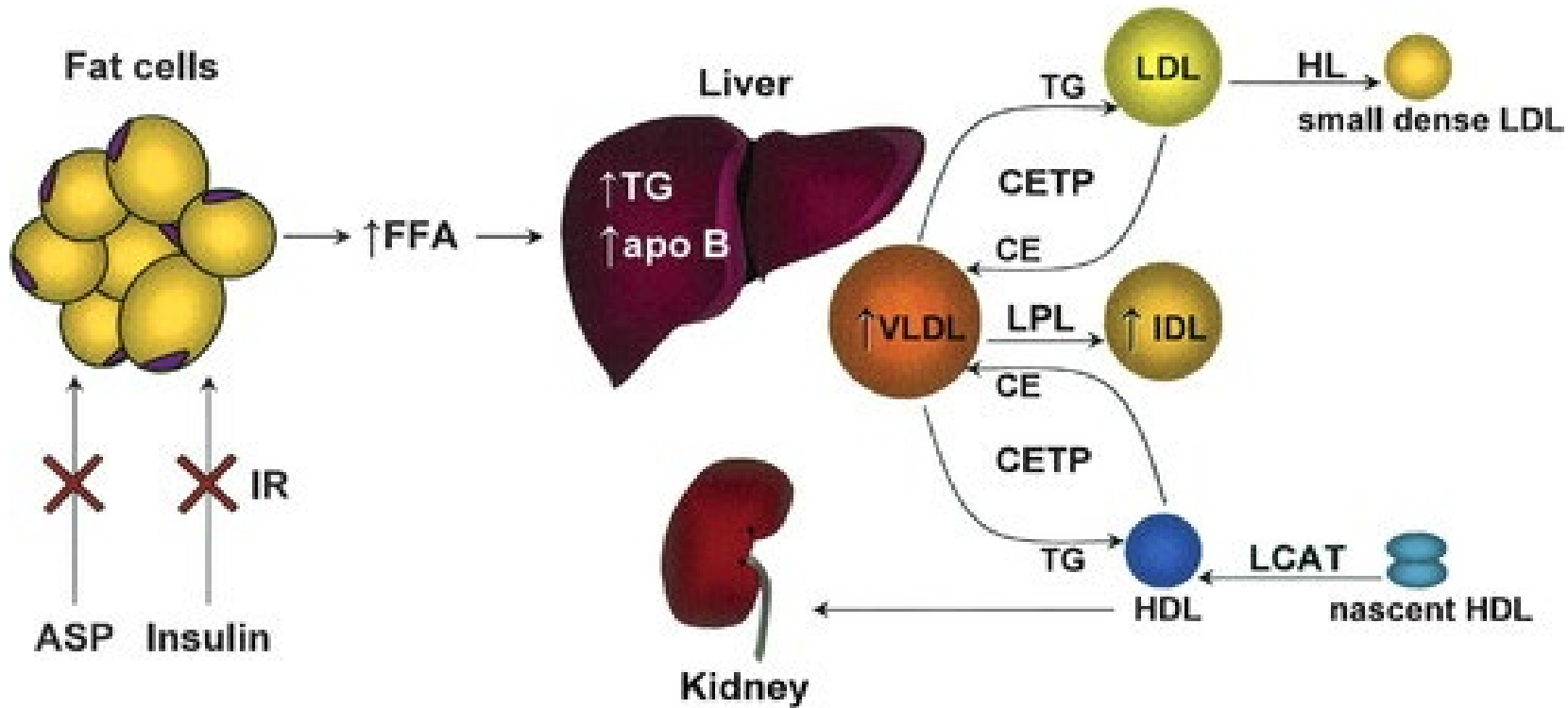
- Azarpazhooh MR, Najafi F, Darbandi M, Kiarasi S, Oduyemi T, Spence JD. Triglyceride/high-density lipoprotein cholesterol ratio: a clue to metabolic syndrome, insulin resistance, and severe atherosclerosis. *Lipids*. 2021;56(4):405-412. doi:10.1002/lipd.12302
- Che B, Zhong C, Zhang R, et al. Triglyceride-glucose index and triglyceride to high-density lipoprotein cholesterol ratio as potential cardiovascular disease risk factors: an analysis of UK biobank data. *Cardiovasc Diabetol*. 2023;22(1):34. doi:10.1186/s12933-023-01762-2
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- Paublini H, López González AA, Busquets-Cortés C, Tomas-Gil P, Riutord-Sbert P, Ramírez-Manent JI. Relationship between atherogenic dyslipidaemia and lipid triad and scales that assess insulin resistance. *Nutrients*. 2023;15(9):2105. doi:10.3390/nu15092105

Lab Abnormalities Associated With Insulin Resistance



- Duez H, Lamarche B, Uffelman KD, Valero R, Cohn JS, Lewis GF. Hyperinsulinemia is associated with increased production rate of intestinal apolipoprotein B-48-containing lipoproteins in humans. *Arterioscler Thromb Vasc Biol.* 2006;26(6):1357-1363. doi:10.1161/01.ATV.0000222015.76038.14
- Eor J, Lee Y, Lee YC, Kwon YJ, Lee JW. Single point insulin sensitivity estimator index is associated with predominance of atherogenic small, dense low-density lipoprotein cholesterol particles in Korean obese adults: a retrospective study. *Korean J Fam Med.* Published online February 20, 2025. doi:10.4082/kjfm.24.0202
- Laakso M, Sarlund H, Mykkanen L. Insulin resistance is associated with lipid and lipoprotein abnormalities in subjects with varying degrees of glucose tolerance. *Arteriosclerosis.* 1990;10(2):223-231. doi:10.1161/01.atv.10.2.223

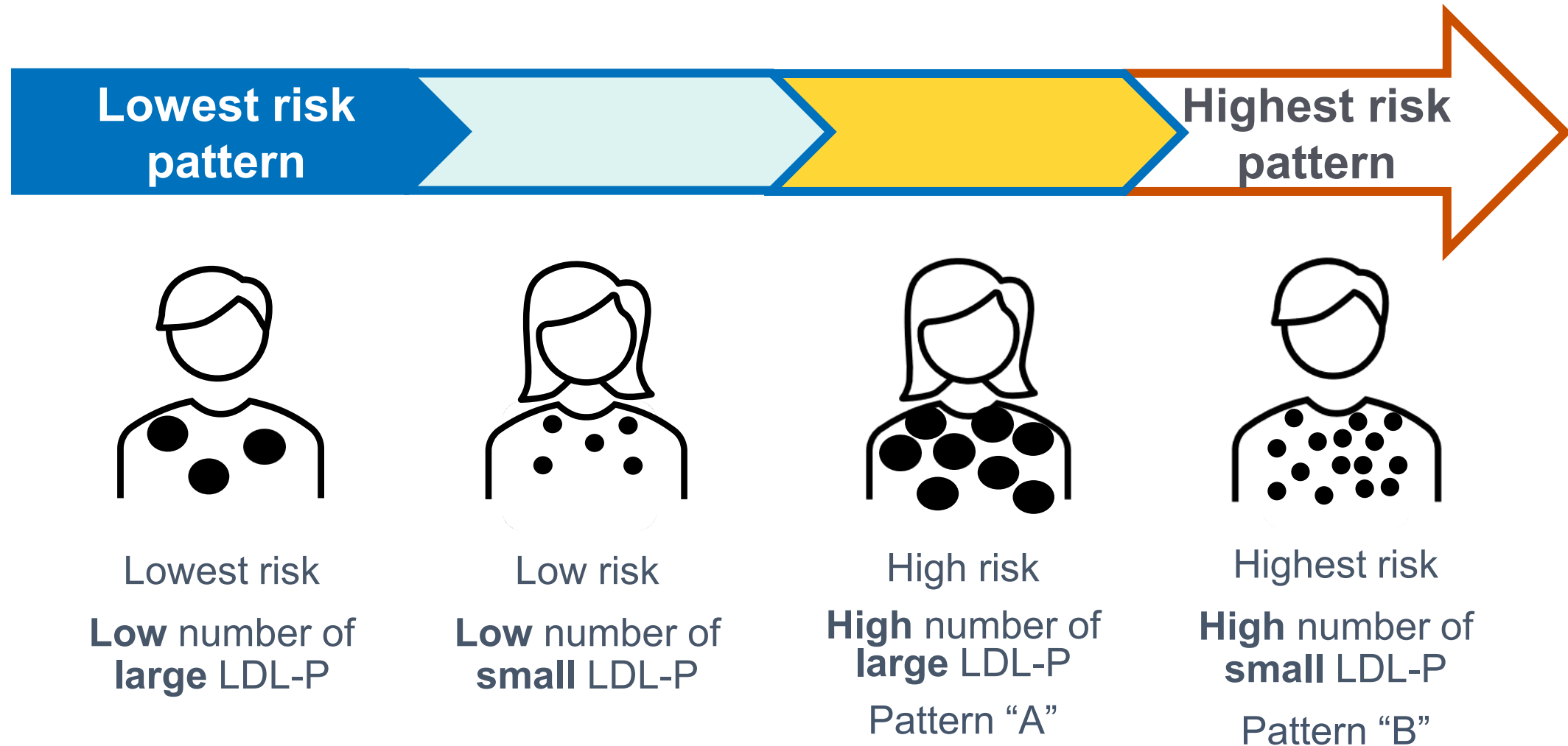
Dyslipidemic Triad



Long Description: Dyslipidemic Triad

The dyslipidemic triad consists of elevated triglycerides, elevated small, dense L D L particles, and low H D L.

LDL Particle (LDL-P) Number & Size



Insulin Resistance and “Triglyceride Release”

Insulin resistance leads to increased release of fatty acids from adipose tissue, which travel to the liver and drive the production of triglyceride-rich VLDL particles.

These VLDL particles then interact with other lipoproteins, such as LDL and HDL, to offload their triglyceride cargo, contributing to the dyslipidemia often observed in insulin-resistant states.

Gamma-Glutamyl Transferase (GGT)

Predictive Biomarker of Diabetes and Insulin Resistance?

```
graph LR; A[Observational studies have shown that elevated GGT levels are associated with an increased risk of T2D.] --> B[The association may be influenced by environmental toxins such as POPs, which contribute to oxidative stress and metabolic disturbances.]; B --> C[Mendelian randomization (MR) studies have not supported a causal effect of GGT on T2D risk.];
```

Observational studies have shown that elevated GGT levels are associated with an increased risk of T2D.

The association may be influenced by environmental toxins such as POPs, which contribute to oxidative stress and metabolic disturbances.

Mendelian randomization (MR) studies have not supported a causal effect of GGT on T2D risk.

References

GAMMA-GLUTAMYL TRANSFERASE (GGT)

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- Koenig G, Seneff S. Gamma-glutamyltransferase: a predictive biomarker of cellular antioxidant inadequacy and disease risk. *Dis Markers*. 2015;2015:818570. doi:10.1155/2015/818570
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- Nannipieri M, Gonzales C, Baldi S, et al. Liver enzymes, the metabolic syndrome, and incident diabetes: the Mexico City diabetes study. *Diabetes Care*. 2005;28(7):1757-1762. doi:10.2337/diacare.28.7.1757
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Clinical Focus

- Antecedents, triggers, and mediators of cardiometabolic syndrome.



Antecedents, Triggers, and Mediators Lead to...



Insulin
resistance

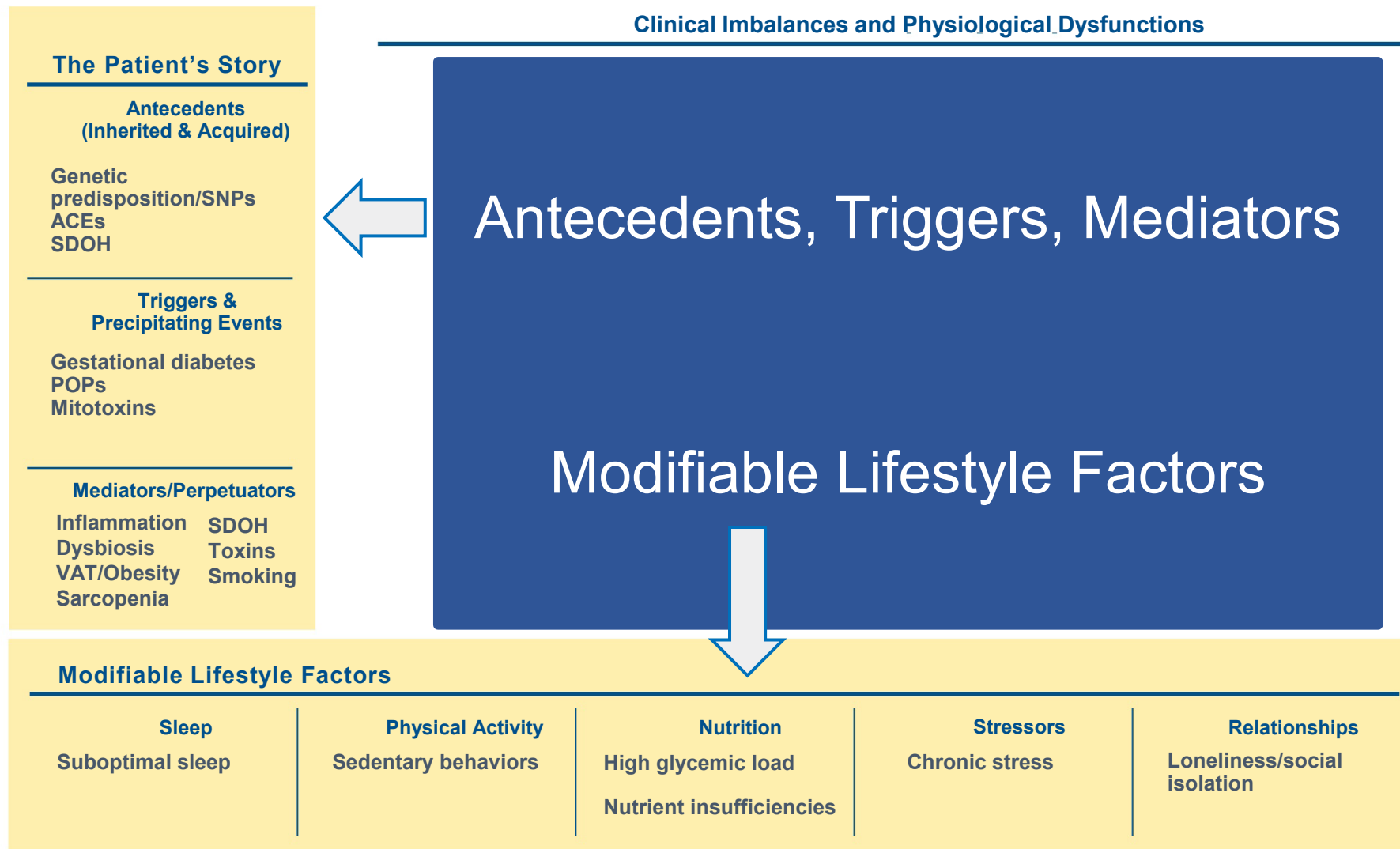


Elevated fasting and
postprandial insulin blood levels



Multi-system dysregulation of
physiological functions

Functional Medicine Matrix



See: "References:
Antecedents, Triggers,
and Mediators of Insulin
Resistance"

Name: _____ Date: _____ CC: _____

Long Description: antecedents and triggers of cardiometabolic disease

- A T M's
 - Antecedents
 - Genetic predisposition / S N P's
 - A C E's
 - S D O H
 - Triggers and precipitating events
 - Gestational diabetes
 - P O P's
 - Mitotoxins
 - Mediators/perpetuators
 - Inflammation
 - S D O H
 - Dysbiosis
 - Toxins
 - V A T / Obesity
 - Smoking
 - Sarcopenia
- Modifiable Lifestyle Factors
 - Sleep
 - Suboptimal sleep
 - Physical Activity
 - Sedentary behaviors
 - Nutrition
 - High glycemic load
 - Nutrient insufficiencies
 - Stressors
 - Chronic stress
 - Relationships
 - Loneliness/social isolation

References

ANTECEDENTS, TRIGGERS, AND MEDIATORS OF INSULIN RESISTANCE

Genetics:

- Brown AE, Walker M. Genetics of insulin resistance and the metabolic syndrome. *Curr Cardiol Rep.* 2016;18(8):75. doi:10.1007/s11886-016-0755-4

Stress:

- Joseph JJ, Golden SH. Cortisol dysregulation: the bidirectional link between stress, depression, and type 2 diabetes mellitus. *Ann N Y Acad Sci.* 2017;1391(1):20-34. doi:10.1111/nyas.13217
- Onyango AN. Cellular stresses and stress responses in the pathogenesis of insulin resistance. *Oxid Med Cell Longev.* 2018;2018:4321714. doi:10.1155/2018/4321714

Microbiome:

- Dabke K, Hendrick G, Devkota S. The gut microbiome and metabolic syndrome. *J Clin Invest.* 2019;129(10):4050-4057. doi:10.1172/JCI129194

Toxins:

- Kim YA, Park JB, Woo MS, Lee SY, Kim HY, Yoo YH. Persistent organic pollutant-mediated insulin resistance. *Int J Environ Res Public Health.* 2019;16(3):448. doi:10.3390/ijerph16030448
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Sarcopenia:

- Cleasby ME, Jamieson PM, Atherton PJ. Insulin resistance and sarcopenia: mechanistic links between common co-morbidities. *J Endocrinol.* 2016;229(2):R67-R81. doi:10.1530/JOE-15-0533

Visceral Adiposity, Inflammation:

- Colleluori G, Perugini J, Giordano A, Cinti S. From obesity to diabetes: the role of the adipose organ. *Handb Exp Pharmacol.* 2022;274:75-92. doi:10.1007/164_2021_572

Nutrients:

- Dubey P, Thakur V, Chattopadhyay M. Role of minerals and trace elements in diabetes and insulin resistance. *Nutrients.* 2020;12(6):1864. doi:10.3390/nu1206186

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ANTECEDENTS, TRIGGERS, AND MEDIATORS OF INSULIN RESISTANCE

Sleep:

- Poggiogalle E, Jamshed H, Peterson CM. Circadian regulation of glucose, lipid, and energy metabolism in humans. *Metabolism*. 2018;84:11-27. doi:10.1016/j.metabol.2017.11.017

Physical Activity:

- Ashcroft SP, Stocks B, Egan B, Zierath JR. Exercise induces tissue-specific adaptations to enhance cardiometabolic health. *Cell Metab*. 2024;36(2):278-300. doi:10.1016/j.cmet.2023.12.008
- Myers J, Kokkinos P, Nyelin E. Physical activity, cardiorespiratory fitness, and the metabolic syndrome. *Nutrients*. 2019;11(7):1652. doi:10.3390/nu11071652

Loneliness:

- Brown EG, Gallagher S, Creaven AM. Loneliness and acute stress reactivity: a systematic review of psychophysiological studies. *Psychophysiology*. 2018;55(5):e13031. doi:10.1111/psyp.13031

Gestational Diabetes:

- Plows JF, Stanley JL, Baker PN, Reynolds CM, Vickers MH. The pathophysiology of gestational diabetes mellitus. *Int J Mol Sci*. 2018;19(11):3342. doi:10.3390/ijms19113342

ACEs/SDOHs:

- Pavlovskaya I, Polcova A, Mechanick JL, et al. Dysglycemia and abnormal adiposity drivers of cardiometabolic-based chronic disease in the Czech population: biological, behavioral, and cultural/social determinants of health. *Nutrients*. 2021;13(7):2338. doi:10.3390/nu13072338
- Wallace C, Krugman R. More than what you eat: a review on the association between childhood maltreatment and elevated adult BMI. *Curr Nutr Rep*. 2024;13(3):377-381. doi:10.1007/s13668-024-00558-4

High Glycemic Load:

- Jung CH, Choi KM. Impact of high-carbohydrate diet on metabolic parameters in patients with type 2 diabetes. *Nutrients*. 2017;9(4):322. doi:10.3390/nu9040322

Psychosocial and socioeconomic conditions are likely important antecedents, triggers, and mediators of perturbations of the HPA axis, which, in genetically susceptible individuals, contribute to visceral adiposity, insulin resistance, and dyslipidemia.

Mediators of Cardiometabolic Dysfunction

Mediator	Effect	Cardiometabolic Parameter Affected
Sleep apnea	Increased	• Inflammation and inflammatory markers (hs-CRP, IL-6) • Insulin resistance • Dyslipidemia
Circadian disruption	Increased	• Impaired hepatic and cellular functions • Obesity • Type 2 diabetes • Metabolic syndrome
Sedentarism	Increased Decreased	• HbA1c • Triglycerides • HDL • ApoA1 • Mitochondrial function
Loneliness & acute stress	Increased Decreased	• Blood pressure • Inflammatory responses • Perception that stressors are more stressful • Cardiac output and heart rate

References

MEDIATORS OF CARDIOMETABOLIC DYSFUNCTION

Circadian Disruption:

- Mukherji A, Bailey SM, Staels B, Baumert TF. The circadian clock and liver function in health and disease. *J Hepatol*. 2019;71(1):200-211. doi:10.1016/j.jhep.2019.03.020

Sleep Apnea:

- Kritikou I, Basta M, Vgontzas AN, et al. Sleep apnoea, sleepiness, inflammation and insulin resistance in middle-aged males and females. *Eur Respir J*. 2014;43(1):145-155. doi:10.1183/09031936.00126712

Lipids:

- Lin X, Zhang X, Guo J, et al. Effects of exercise training on cardiorespiratory fitness and biomarkers of cardiometabolic health: a systematic review and meta-analysis of randomized controlled trials. *J Am Heart Assoc*. 2015;4(7):e002014. doi:10.1161/JAHA.115.002014
- Nadeem R, Singh M, Nida M, et al. Effect of obstructive sleep apnea hypopnea syndrome on lipid profile: a meta-regression analysis. *J Clin Sleep Med*. 2014;10(5):475-489. doi:10.5664/jcsm.3690

Loneliness/Stress:

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- Spruill TM, Butler MJ, Thomas SJ, et al. Association between high perceived stress over time and incident hypertension in Black adults: findings from the Jackson Heart Study. *J Am Heart Assoc*. 2019;8(21):e012139. doi:10.1161/JAHA.119.012139

References

MEDIATORS OF CARDIOMETABOLIC DYSFUNCTION

Physical Activity/Exercise:

- Fiorenza M, Lemminger AK, Marker M, et al. High-intensity exercise training enhances mitochondrial oxidative phosphorylation efficiency in a temperature-dependent manner in human skeletal muscle: implications for exercise performance. *FASEB J*. 2019;33(8):8976-8989. doi:10.1096/fj.201900106RRR
- Pan B, Ge L, Xun YQ, et al. Exercise training modalities in patients with type 2 diabetes mellitus: a systematic review and network meta-analysis. *Int J Behav Nutr Phys Act*. 2018;15(1):72. doi:10.1186/s12966-018-0703-3

IFM Toolkit Items



Lifelong Lifestyle Timeline

	Birth	Current Age
Sleep		
Exercise		
Nutrition		
Stress		
Relationships		

Current Concerns: _____

Name: _____ Date: _____



Lifestyle Questionnaire

Patient Name: _____

Date: _____

Sleep

How many hours of sleep do you get each night on average: _____

Do you have problems falling asleep?

☐ Yes

☐ No

Do you have problems staying asleep?

☐ Yes

☐ No

Do you have problems with insomnia?

☐ Yes

☐ No

Do you snore?

☐ Yes

☐ No

Do you feel rested upon awakening?

☐ Yes

☐ No

Do you use sleeping aids?

☐ Yes

☐ No

If yes, explain: _____

Physical Activity

List your current physical activity, including type, number of times per week, and duration in minutes:

Cardiovascular/Aerobic (e.g., brisk walk): _____

Strength/Resistance: _____

Flexibility/Stretching: _____

Balance: _____

Sports/Leisure (e.g., golf): _____

Other (e.g., yard work): _____

Clinical Focus

- Pathophysiological mechanisms of cardiometabolic syndrome.



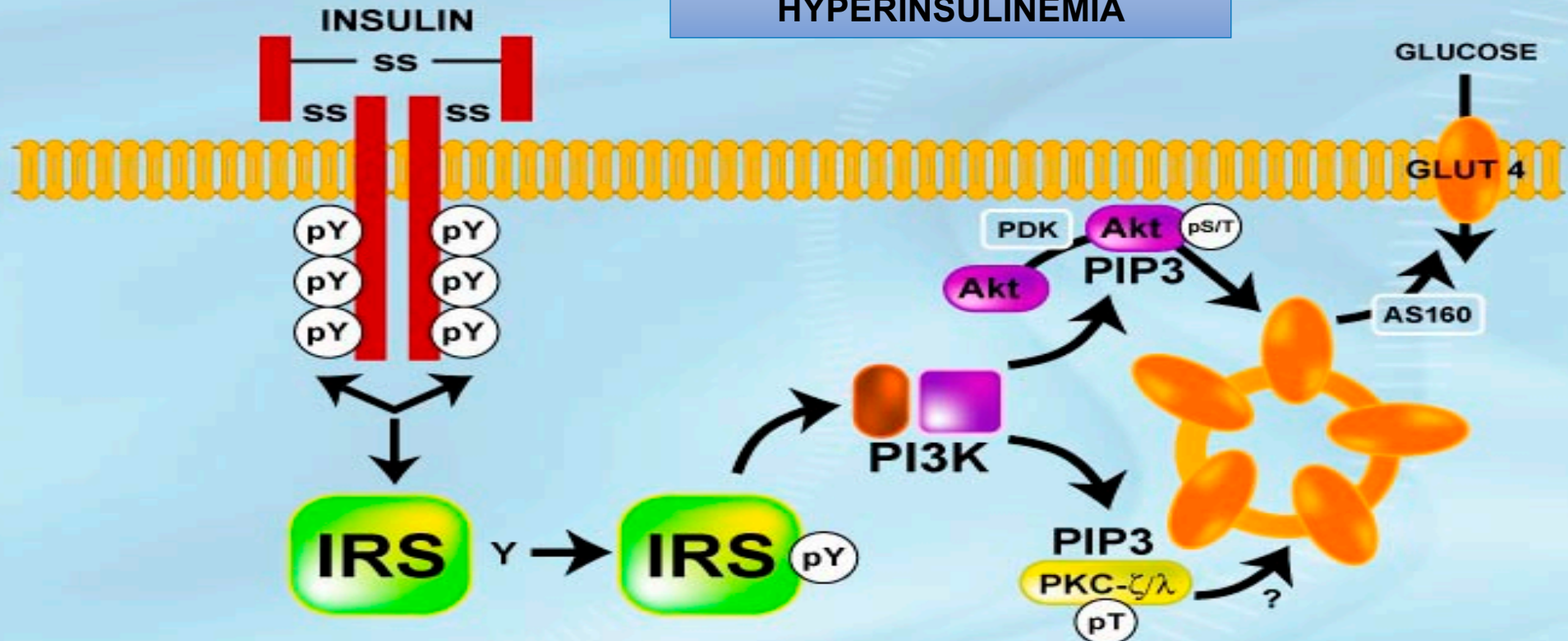
Insulin Resistance and Hyperinsulinemia

- The ability of insulin to mediate glucose uptake into the cell can vary six to seven-fold between persons.
- This variation is a consequence of cellular insensitivity to the action of insulin and is termed insulin resistance.
- If able to, the pancreas will compensate by secreting large amounts of insulin to adequately dispose of glucose, resulting in a fasting and postprandial hyperinsulinemic state.

SIMPLIFIED SCHEME OF INSULIN ACTION ON GLUCOSE TRANSPORT



HYPERINSULINEMIA



Legend

Akt= protein kinase

AS160 = Akt substrate of 160 kDa

GLUT = glucose transporter

IRS = insulin receptor substrate-1/2

PDK = phosphoinositide-dependent protein kinase

PIP3 = phosphatidylinositol 3 triphosphate

PI3K= phosphatidylinositol [3,4,5) kinase

PKC = protein kinase C

pS/T = serine/threonine phosphorylation

pT = threonine phosphorylation

pY = tyrosine phosphorylation

Source: International Chair on Cardiometabolic Risk
www.cardiometabolic-risk.org

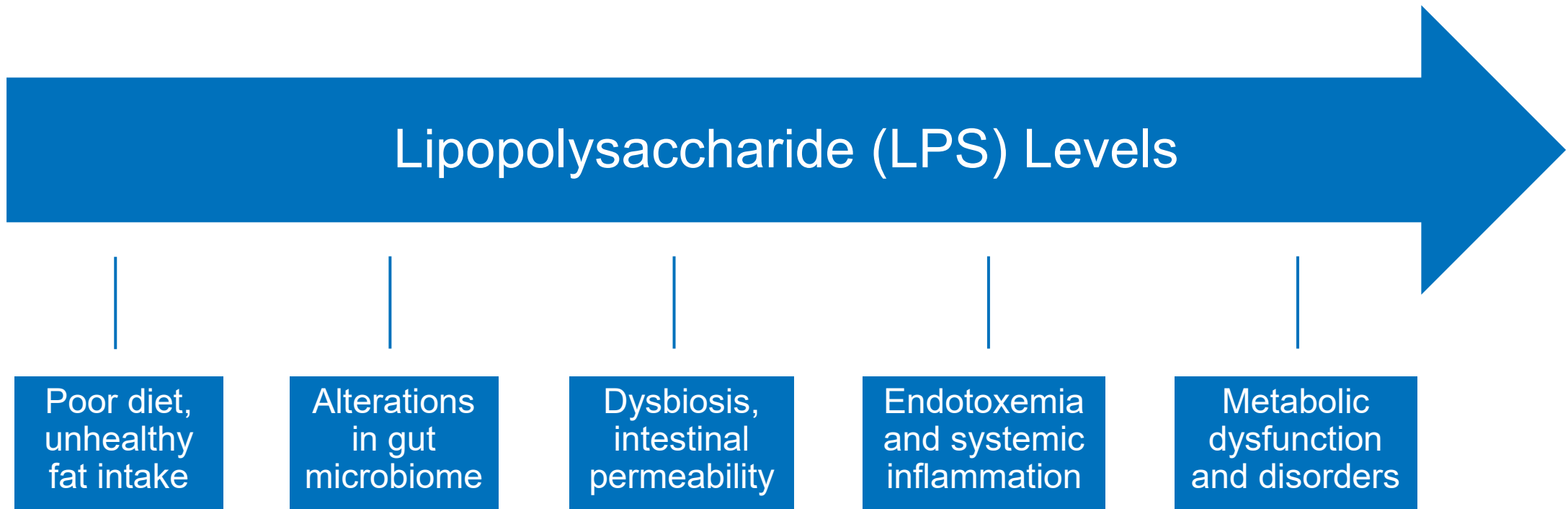
Long Description: Glucose Transport

Simplified scheme of insulin action on glucose transport

During healthy insulin metabolism, insulin binds to a cell membrane receptor, initiating a signal transduction cascade of protein kinases. This process results in glucose entering the cell via the membrane protein GLUT 4 transporter. This process is dysregulated in the presence of hyperinsulinemia, impairing the transport of glucose through GLUT 4 into the cell.

Insulin Resistance

Nutrition, GI Dysfunction, and Endotoxemia



- Cândido FG, Valente FX, Grześkowiak ŁM, Moreira APB, Rocha DMUP, Alfenas RCG. Impact of dietary fat on gut microbiota and low-grade systemic inflammation: mechanisms and clinical implications on obesity. *Int J Food Sci Nutr*. 2018;69(2):125-143. doi:10.1080/09637486.2017.1343286
- Fuke N, Nagata N, Suganuma H, Ota T. Regulation of gut microbiota and metabolic with dietary factors. *Nutrients*. 2019;11(10):2277. doi:10.3390/nu11102277

Insulin Resistance and Endotoxemia

- Endotoxemia decreases insulin sensitivity and induces insulin resistance, without an effect on pancreatic beta-cell function.
- Acute inflammation induces systemic IR in humans, following modulation of specific adipose inflammatory and insulin signaling pathways.

- Mehta NN, McGillicuddy FC, Anderson PD, et al. Experimental endotoxemia induces adipose inflammation and insulin resistance in humans. *Diabetes*. 2010;59(1):172-181. doi:10.2337/db09-0367
- Patel PN, Shah RY, Ferguson JF, Reilly MP. Human experimental endotoxemia in modeling the pathophysiology, genomics, and therapeutics of innate immunity in complex cardiometabolic diseases. *Arterioscler Thromb Vasc Biol*. 2015;35(3):525-534. doi:10.1161/ATVBAHA.114.304455
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- Zu L, He J, Jiang H, Xu C, Pu S, Xu G. Bacterial endotoxin stimulates adipose lipolysis via toll-like receptor 4 and extracellular signal-regulated kinase pathway. *J Biol Chem*. 2009;284(9):5915-5926. doi:10.1074/jbc.M807852200

Insulin Resistance

Stress, Obesity, and Inflammation



Stress, primarily through hyperactivation of the HPA axis, appears to contribute to the development of visceral adiposity, and vice versa.



Obesity itself seems to constitute a chronic stressful state and may cause HPA axis dysfunction. Obesity may be considered a systemic, low-grade inflammatory condition.



Recent data implicate several neuroendocrine and inflammatory mechanisms mobilized during chronic stress in the development of central obesity and metabolic syndrome.



Psychosocial and socioeconomic stressors are important triggers for the perturbations of the HPA axis, which, in genetically susceptible individuals, are followed by visceral obesity, insulin resistance, and dyslipidemia.

- Kyrou I, Chrousos GP, Tsigos C. Stress, visceral obesity, and metabolic complications. *Ann N Y Acad Sci.* 2006;1083:77-110. doi:10.1196/annals.1367.008
- Joseph JJ, Golden SH. Cortisol dysregulation: the bidirectional link between stress, depression, and type 2 diabetes mellitus. *Ann N Y Acad Sci.* 2017;1391(1):20-34. doi:10.1111/nyas.13217
- Rosmond R. Stress induced disturbances of the HPA axis: a pathway to type 2 diabetes? *Med Sci Monit.* 2003;9(2):RA35-RA39.

Social Determinants of Health and Adverse Childhood Experiences (ACEs)



Factors such as no or limited education, poor health literacy and access to healthcare, low income, poor eating patterns, and high alcohol consumption can lead to dysglycemia- and adiposity-related complications.



Adverse childhood experiences (ACEs), childhood maltreatment, and abuse have been linked to cardiometabolic comorbidities, adulthood diabetes, insulin resistance, and obesity. Gender and racial disparities have not yet been fully elucidated.



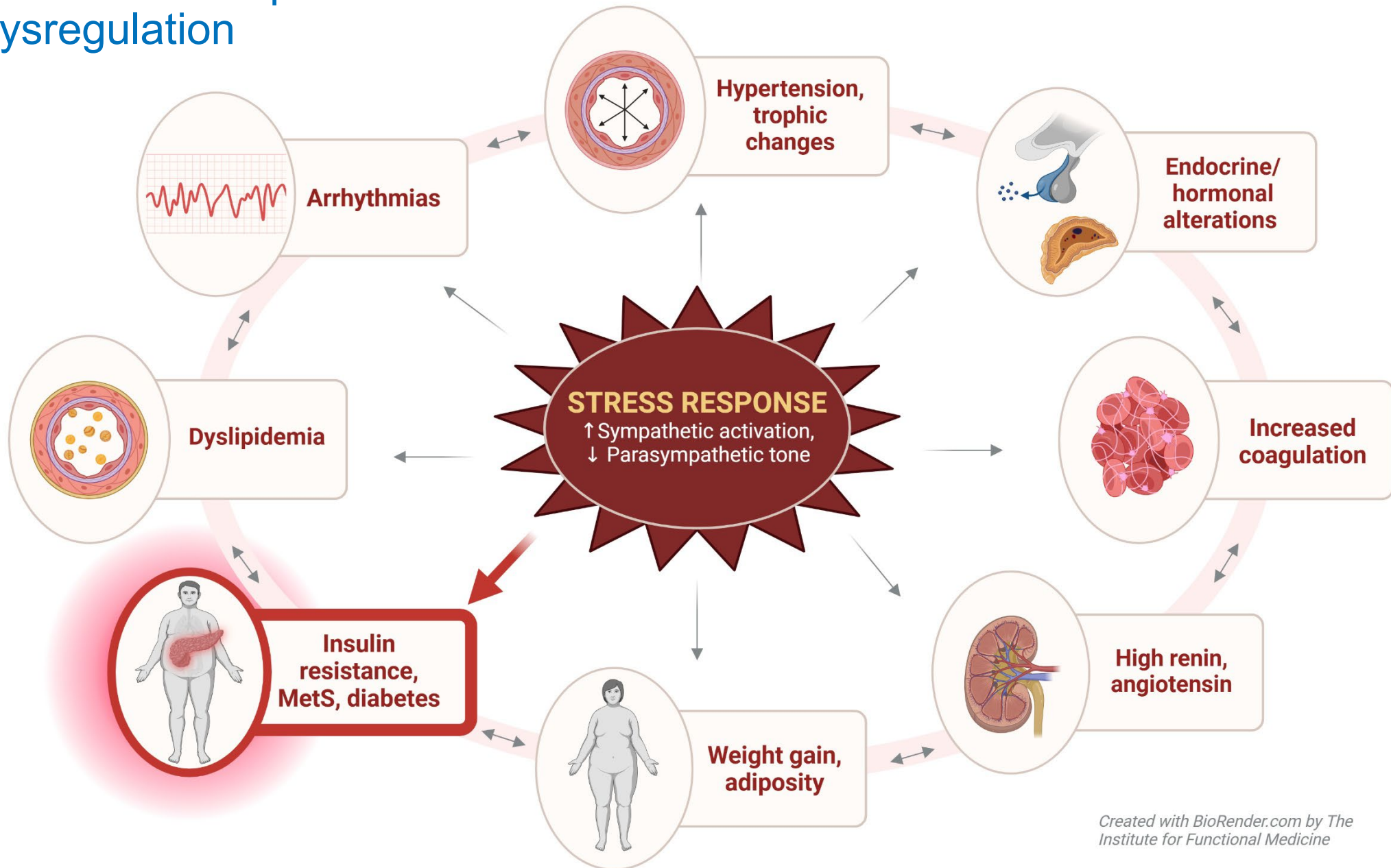
Clinicians should gather information from patients about childhood traumas and/or abuse and SDOH in a sensitive, trauma-informed manner.

References

SOCIAL DETERMINANTS OF HEALTH AND ADVERSE CHILDHOOD EXPERIENCES (ACES)

- Chen Z, Zhang K, Peng S, et al. Climate change and air pollution can amplify vulnerability of glucose metabolism: the mediating effects of biological aging. *Environ Res*. 2025;272:121183. doi:10.1016/j.envres.2025.121183
- Onyango AN. Cellular stresses and stress responses in the pathogenesis of insulin resistance. *Oxid Med Cell Longev*. 2018;2018:4321714. doi:10.1155/2018/4321714
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The Stress Response as Mediator of Cardiovascular and Cardiometabolic Dysregulation



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Institute for Functional Medicine

Long Description: The Stress Response as Mediator of Cardiovascular and Cardiometabolic Dysregulation

The stress response involves increased sympathetic activation and decreased parasympathetic tone leading to:

- Insulin resistance, metabolic syndrome, diabetes
- Dyslipidemia
- Arrhythmias
- Hypertension, trophic changes
- Endocrine and hormonal alterations
- Increased coagulation
- High renin, angiotensin
- Weight gain, adiposity

Insulin Resistance and Persistent Organic Pollutants

POPs have a long half-life and can exist for years or decades in the air, water, food supply, and living organisms.

POPs are believed to:

- Alter glucose transport activity
- Disrupt the insulin signaling pathway
- Increase adiposity (POPs are lipophilic)
- Impair the immune, reproductive, and nervous systems

Evidence suggests exposure to low levels of POPs can increase the risk of insulin resistance and metabolic disorders.

References

INSULIN RESISTANCE AND PERSISTENT ORGANIC POLLUTANTS

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Insulin Resistance and Mitochondrial Dysfunction

Mitotoxigants (toxins, medications):

- Reduce mitochondrial membrane integrity and ATP production
- Increase lipids in the cytoplasm mediated by AMPK
- Lead to poor glucose uptake
- Impair insulin signaling pathway
- Increase levels of reactive oxygen species (ROS) and malonaldehyde (MDA)

The increased oxidative stress can lead to insulin resistance in skeletal muscle.

Transport Processes and Cardiometabolic Dysfunction: Part 3

The Functional Medicine Matrix

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Foundational lifestyle therapeutic interventions for cardiometabolic syndrome.



Foundational Therapeutic Approach

Cardiometabolic Dysfunction



LIFESTYLE FIRST



FOOD FIRST



GUT FIRST

Getting Healthy Is a Team Sport



Foundational Therapeutic Interventions

- **Referrals that address ATMs; address the matrix**
- **Lifestyle modification**
- Dietary modifications
- Gut health
- Foundational supplements – to increase cellular responsiveness to insulin (i.e., sensitivity)

Address the Mediators

Modifiable Lifestyle Factors

MODIFIABLE LIFESTYLE FACTORS

Modifiable Lifestyle Factors				
Sleep 	Physical Activity	Nutrition	Stressors	Relationships

Sleep and Circadian Rhythms

Effects on Glucose, Lipid, and Energy Metabolism

- Disrupted circadian rhythms and sleep are associated with:
 - Increased risk of metabolic diseases
 - Elevated levels of glucose, insulin, and triglycerides
 - Decreased beta cell function and energy expenditure
- **Circadian alignment factors include:**
 - Sufficient daytime light exposure
 - Food intake during the day (not at night)
 - Sleeping during the biological night (even consistent daytime sleep patterns cause misalignment)

IFM Toolkit Item

- Avoid electronic devices at least two hours before bed.
- Sleep in a cool bedroom.
- Avoid napping late in the day.
- And more...



Suggestions for Better Sleep

Achieving deep, refreshing sleep isn't always as simple as cozying up in bed and closing your eyes – it often depends on your daily lifestyle choices and habits. From how you unwind before bed to what you eat, small adjustments can make a big difference in your sleep. This guide walks you through suggestions that can help you achieve better sleep, so you can wake up feeling well-rested, energetic, and ready to start your day.

Step 1: Tune Up Your Sleep Hygiene

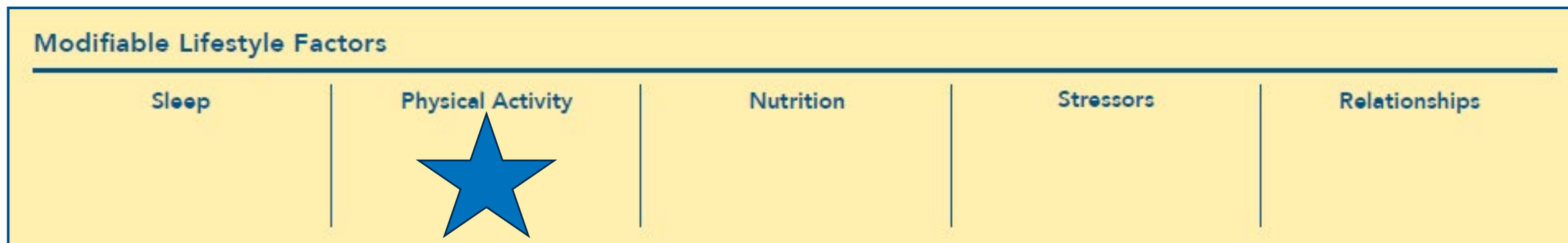
Sleep hygiene is a set of healthy habits and environmental conditions that can help you sleep soundly. If you are struggling with your sleep, tuning up your sleep hygiene is the place to start. Try the following sleep hygiene tune-ups for better sleep:

- **Go to bed and wake up at the same time daily.** A consistent waking time and bedtime each day supports your circadian rhythm, your body's internal 24-hour clock that regulates sleep and other body functions. A healthy circadian rhythm helps you stay alert and energetic during the day and eases you to sleep at night.¹ Ideally, your bedtime and wake time should be set to allow for 7-9 hours of sleep each night.²
- **Sleep in a dark bedroom.** Darkness promotes sleep by supporting a healthy circadian rhythm. Even dim artificial light exposure at night from sources like streetlights can interfere with melatonin, a hormone that regulates sleep.^{3,4} If you have lights outside your bedroom at night or work the night shift and sleep during the day, hang blackout curtains on your bedroom windows or wear an eye mask. Remove electronics that have bright lights from your bedroom. For example, replace a glowing digital alarm clock with a non-digital alarm clock.
- **Avoid spending time on electronic devices two hours before bed.**^{5,6} The bright light from electronic devices such as smartphones and tablets can increase alertness and reduce sleepiness, making it harder to drift off to sleep.^{5,7} Pick an activity that doesn't involve electronic devices before bed. For example, read a hardcover or paperback book.⁸⁻¹⁰ If you must use electronic devices like eReaders close to bedtime, put the device on "nighttime mode." Wearing blue light-blocking glasses while using electronic devices at night may reduce the negative impact of these devices on sleep.¹¹

Address the Mediators

Modifiable Lifestyle Factors

MODIFIABLE LIFESTYLE FACTORS



Physical Activity

Improved Parameters for Cardiometabolic Syndrome

- Decreased SBP and DBP by an average of 5-7 mmHg.
 - Reductions of 2 mmHg SBP and DBP reduce the risk of CAD by up to 9%.
- Smokers who become physically active are more likely to stop smoking.
- Improves insulin sensitivity, beta-cell function, GLP-1 sensitivity.
- Reduces visceral adipose tissue and lipid deposition.

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• Ghadieh AS, Saab B. Evidence for exercise training in the management of hypertension in adults. *Can Fam Physician.* 2015;61(3):233-239.

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Four Parts of an Exercise Rx

Frequency
Intensity
Time
Type

Accessible version of this resource in your toolkit.



Exercise Prescription

Name Birthdate

Goals

Risk Assessment

- ☐ PAR-Q: ☐ Cleared ☐ Not cleared
☐ Risk Factors: ☐ Low Risk ☐ Moderate Risk ☐ High Risk
☐ Exercise Stress Test: ☐ Pass ☐ Fail

Comments:

Intensity (Check desired measure type and intensity level):

☐ "Talk Test"	☐ Perceived Exertion (10 Point Scale)	☐ Maximal Heart Rate (HR _{max})* =
☐ Low intensity (able to talk and/or sing)	☐ Low (less than 3)	☐ Low (less than 64% =)
☐ Moderate intensity (able to talk but not sing)	☐ Moderate (3 to 4)	☐ Moderate (64% to 76% =)
☐ Vigorous intensity (difficult or unable to talk)	☐ Vigorous (more than or equal to 5)	☐ Vigorous (more than 76% =)

*Calculating HR_{max} = 206.9 - (0.67 x age)

Comments:

FITT Plan

Exercise Prescription	Cardio/Aerobic	Strength/Resistance	Flexibility/Stretching	Balance
F — Frequency Times per week				
I — Intensity (e.g., low, moderate, vigorous)				
T — Time/duration Minutes each day				
T — Type (e.g., walking, jogging, swimming)				

These recommendations should be followed under the supervision and guidance of a qualified healthcare professional.

Comments:

Recommended by Date

Reassessment Appointment

Based on the American College of Sports Medicine's Exercise is Medicine: A Clinician's Guide to Exercise Prescription. Philadelphia (PA): Lippincott Williams & Wilkins; 2009: 99-133.

Frequency, Intensity, Time, Type (FITT)

PHYSICAL ACTIVITY IN INDIVIDUALS WITH T2DM: A CONSENSUS STATEMENT FROM ACSM

Aerobic

- 150-300 min/week total
- 30-60 min moderate intensity 5x/week, or 20-60 min vigorous intensity 3x/week
- >10 min sessions per session acceptable

Resistance

- Train each muscle group 2-3x/week, not on consecutive days

Flexibility

- 10-30s/stretch per muscle group, 2-4 repetitions, 2-3x/week

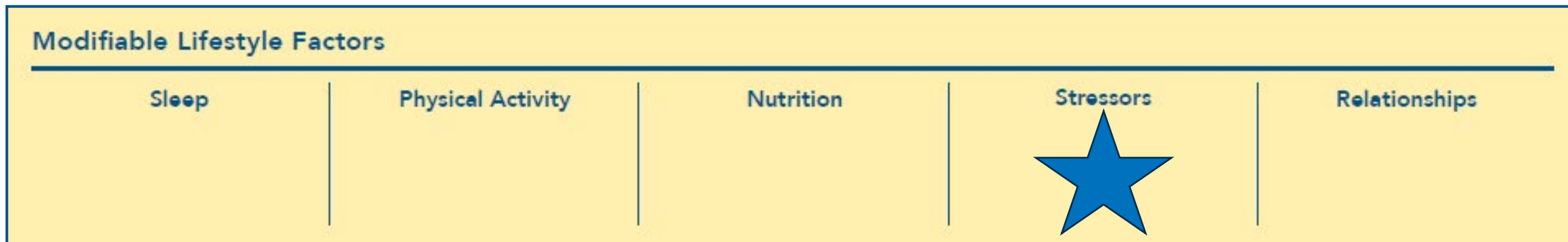
Balance

- Balance, agility, coordination, gait
- 2-3x/week, no set duration

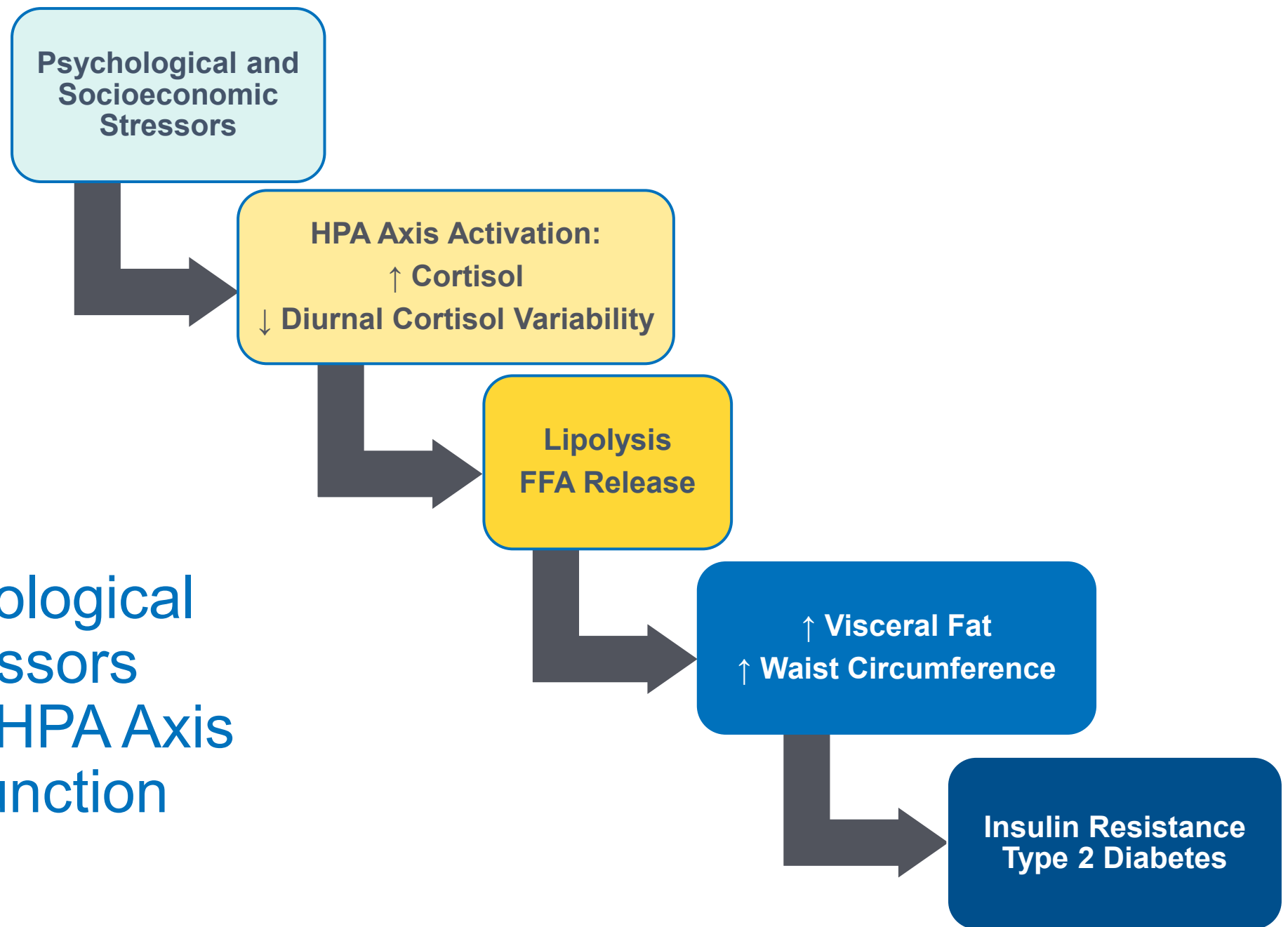
Address the Mediators

Modifiable Lifestyle Factors

MODIFIABLE LIFESTYLE FACTORS



Psychological
Stressors
Trigger HPA Axis
Dysfunction



IFM Toolkit Items

- Breathing techniques
- Mindful walking
- Guided imagery
- And more...



Integrating Guided Imagery and Visualization to Achieve Personal Goals



Mindful Breathing

Mindful breathing is a simple tool that can easily be used anywhere to support your health and well-being.^{1,2} While it may take some practice, anyone can begin by slowing down their breath and focusing on each inhale and exhale. With regular practice, intentional and mindful breathing can help strengthen the mind-body connection which has physical, mental, and emotional benefits. With time, it may result in benefits such as supporting heart health,³ better sleep,⁴ stress,³ and improved mental health.^{5,6}

What is Mindful Breathing

Breathing is an activity most people do without even thinking about it. Since it is done automatically, most people aren't paying attention to whether their breathing is shallow or deep. Shallow breathing comes from the chest, bringing only small amounts of air into the lungs. Deep breathing pulls more air into the lungs, which requires the use of a muscle called the diaphragm. When inhaling the diaphragm muscle contracts and when exhaling, the diaphragm muscle relaxes. Deep, diaphragmatic, also known as abdominal breathing encourages deep, slow breaths that activate the relaxation centers in the brain.⁷⁻⁹

Mindful breathing exercises utilize deep, diaphragmatic breathing. They are designed to help you focus on inhaling and exhaling in a deep, controlled manner. When you incorporate mindful abdominal breathing exercises into your daily routine, it allows you to shift your attention away from your current situation and instead focus on your breathing. This can be particularly helpful during stressful or uncomfortable moments. Mindful, deep breathing is an important skill that can help you deal with stress^{9,10,11} and anxiety.¹²⁻¹⁴

How Mindful Breathing Supports Your Health

Intentionally and mindfully slowing down your breath has a direct influence on your autonomic nervous system (ANS) which controls the automatic functions of your body, like heart rate, digestion, and blood pressure.¹⁵ The ANS has two parts:

- The parasympathetic nervous system, also known as the "rest and digest" response
- The sympathetic nervous system, also known as the "fight or flight" response

The goal is to activate the parasympathetic (PS) nervous system which promotes the relaxation response.¹⁴ Mindful breathing helps shift your body out of "fight or flight" and into "rest and digest" by turning on the PS nervous system and turning off the sympathetic nervous system. Once you feel comfortable with your ability to breathe into the abdomen, it is not necessary to use your hands on your abdomen and chest.

Guided imagery and visualization are mind-body therapies that can help improve your health and well-being. These imagery techniques focus on using sight, sound, touch, smell) to make imagined scenarios feel real. Imagining specific scenes or sequences of events may help you achieve your health goals.

Guided imagery and visualization

Guided imagery involves visualizing scenarios or sequences to reduce stress, promote mental and emotional well-being. It may be led by a therapist, or an audio recording previously recorded by a therapist. Guided imagery is immersive. It focuses on scenes in detail and incorporates as many senses (sight, sound, touch, smell, taste) as possible.¹

Unlike visualization, guided imagery, though it differs in several ways. It focuses on something in your mind, such as a health goal you want to achieve. With guided imagery, it may focus on imagining scenes in as much detail as possible. You can also use visualization to imagine performing a series of actions to achieve a goal. It can be either guided or self-directed. Self-directed visualization allows you to choose what scenes to imagine or what steps to take to achieve your goal, without input from external audio suggestions. While guided imagery focuses on visuals, you can also incorporate other senses.²

Visualization

Visualization, like guided imagery and visualization, can help promote focus, preparation and motivation toward a task, behavior, or health goal.

Visualization is beneficial for relaxation^{4,5} and stress relief,^{4,7} and has many health benefits. These techniques also have positive effects on factors that contribute to overall health. Some of these benefits include deep sleep,^{4,8} food consumption^{4,9} and cravings,^{4,10,11} physical activity regulation,^{4,12-15} self-forgiveness,^{4,14,17} and even alcohol consumption cessation.^{4,19}

Address the Mediators

Modifiable Lifestyle Factors

MODIFIABLE LIFESTYLE FACTORS

Modifiable Lifestyle Factors				
Sleep	Physical Activity	Nutrition	Stressors	Relationships 

Loneliness and Cardiometabolic Risk

- Greater levels of loneliness observed to be predictive of exaggerated blood pressure and inflammation responses to acute stress.
- Systematic review: loneliness associated with higher blood pressure at baseline.
- Lonely individuals also have lower cardiac output and heart rate compared to non-lonely people, indicating that loneliness may cause a blunted response to stress compared to non-lonely people.
- Lonely individuals may perceive certain stressors as more stressful than others.



IFM Toolkit Item

- Volunteer
- Take a class and learn a new skill
- Adopt a pet or take care of a neighbor's pet
- And more...

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Importance of Social Support & Relationships

Social interaction and good relationships can help you feel well and live a healthy balanced life. When these key contributors to health are lacking, the risk of disease increases.¹ The following diagram shows some of the health effects associated with poor social support, isolation, and loneliness.^{2,3}

The diagram shows a human silhouette with arrows pointing to various organs and systems, each with a list of health effects:

- Brain & Nervous System:**
 - Decreased brain function and memory⁴
 - Higher risk of dementia (e.g., Alzheimer's)^{5,6}
 - Poorer coping with chronic disease⁷
 - Greater risk for depression and anxiety⁸
- Heart & Blood Vessels:**
 - Greater risk of high blood pressure, heart disease, and stroke⁹
 - More likely to have excess abdominal fat, a risk factor for heart disease¹⁰
 - Greater risk of unhealthy cholesterol levels¹¹
- Gut Health & Digestion:**
 - Greater risk of gut issues like heartburn, bloating, nausea, and constipation¹²
 - Worse symptoms in chronic gut diseases (e.g., Crohn's)^{13,14}
 - Changes in healthy gut bacteria¹⁵
- Blood Sugar Control:**
 - Higher blood sugar levels¹⁶
 - Greater risk of type 2 diabetes¹⁷
 - Poorer diabetes management¹⁸
- Musculoskeletal:**
 - Weaker (less dense) bones¹⁹ and bone issues (fractures and bone mineral density)¹⁴⁻¹⁶ in elderly population^{17,18}
 - Higher risk of muscle wasting^{19,20}
- Inflammation & Immunity:**
 - Increased inflammation^{21,22}
 - Slower wound healing²³
 - Weakened immunity²⁴
- Energy & Cell Health:**
 - Defective mitochondria (the energy factories in most cells)²⁵
 - Greater risk of cellular damage and dysfunction²⁶
 - Faster aging²⁷
- Hormone Regulation:**
 - Too much or too little of the "stress hormone" cortisol²⁸
 - Greater risk of low thyroid function²⁹
 - Altered sex hormone levels³⁰

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Version 3; updated 2023

Tips to Increase Your Support Network

Building and keeping a good social support network may take a little more effort of your life, such as when you move to a new town or retire. Here are some strategies you can use to expand your network of close friends, family, and mentors.

Types of relationship types: This could include both local and long-distance. For example, perhaps there is someone in your local community who is on the same sports team, while a long-distance friend shares a common hobby with you. Each person can provide a different aspect of support.

Virtual connection: Schedule virtual or phone dates with long-distance friends. You can also walk with local friends.

Volunteering: For example, consider volunteering at a charity, food bank, animal shelter, or community organization.

Meetup groups: Many areas have a wide variety of interest-based activity groups like outdoor or mall walking. There are also online options, such as those learning a new language.

Classes: If you are focused on something you'd like to learn more about, taking a class or joining a group is a great way to interact with new people and learn new skills.

Travel: A bus tour or other trip organized for a group could be a fun way to travel if you enjoy travelling.

Helping others: Offering to lend a helping hand is a great way to start a relationship with your neighbors. For example, let them know you can help them move or put out their trash bins when they travel.

Pets: Pets can provide positive mental and physical health benefits. Petting a cat or walking your dog can lower levels of the stress hormone cortisol.³¹ If you don't have a pet, consider walking a neighbor's dog or adopting a pet.³²⁻³⁴

Version 3; updated 2023

Overall Lifestyle Changes



Transport Processes and Cardiometabolic Dysfunction: Part 4

The Functional Medicine Matrix

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Foundational dietary therapeutic interventions for cardiometabolic syndrome.

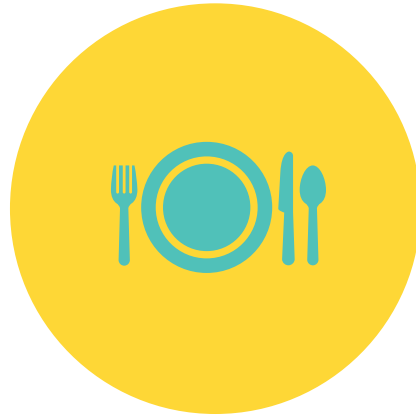


Foundational Therapeutic Approach

Transport Processes and Cardiometabolic Dysfunction



LIFESTYLE FIRST

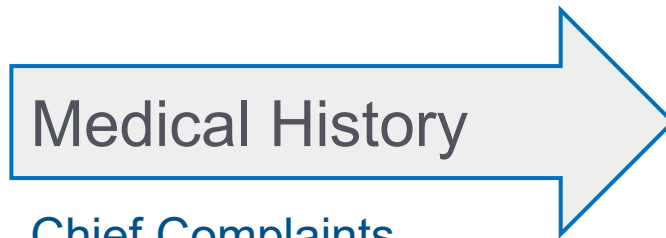


FOOD FIRST



GUT FIRST

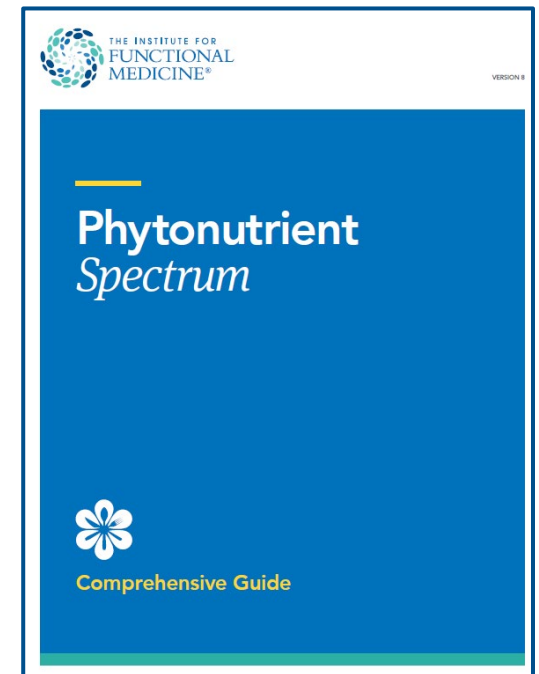
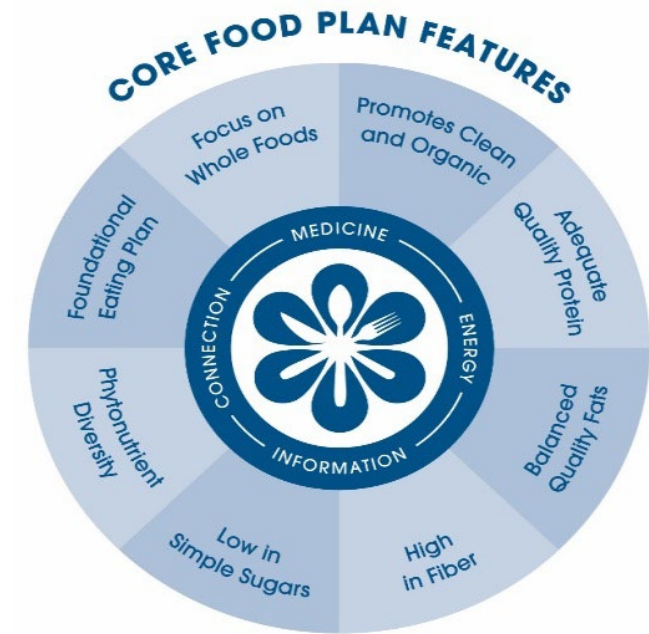
The Clinical Path to the Foundational Food Plans



Chief Complaints
Conditions
Timeline and ATMs
Medication Review



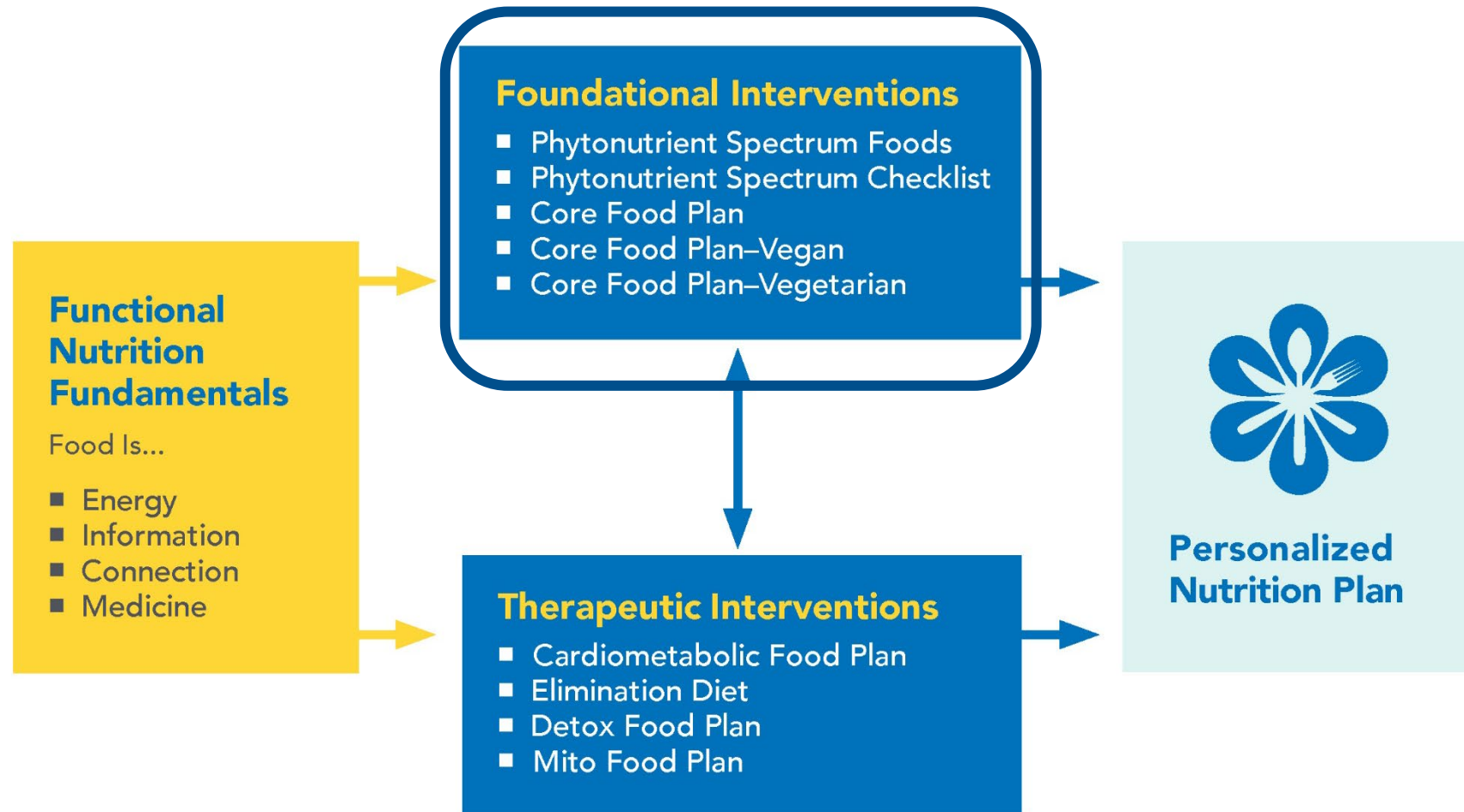
Anthropometrics
Biomarkers and Labs
Clinical Indications From NPE
Diet and Lifestyle Review
Matrix Review



Long Description: functional nutrition food plans

- Medical history
 - Chief complaints
 - Conditions
 - Timeline and A T Ms
 - Medication review
- A B C Ds of Nutrition Evaluation
 - Anthropometrics
 - Biomarkers and labs
 - Clinical indications from N P E
 - Diet and lifestyle review
 - Matrix review
- Core Food Plan
- Phytonutrient spectrum

Nutrition Interventions



Long Description: Nutritional interventions

- Nutrition interventions
 - Use the functional nutrition fundamentals, foundational interventions, and therapeutic interventions to create a personalized nutrition plan.
- Functional Nutrition Fundamentals
 - Food is energy, information, connection, and medicine
- Foundational Interventions
 - Phytonutrient spectrum foods
 - Phytonutrient spectrum checklist
 - Core food plan
 - Core food plan vegan
 - Core food plan vegetarian
- Therapeutic Interventions
 - Cardiometabolic food plan
 - Elimination diet
 - Detox food plan
 - Mito food plan

Nutritional Principles for Insulin Resistance

- Decrease insulin stimulation
- Increase cellular responsiveness to insulin (sensitivity)

Modulation of Insulin Secretion

Macronutrient Selection

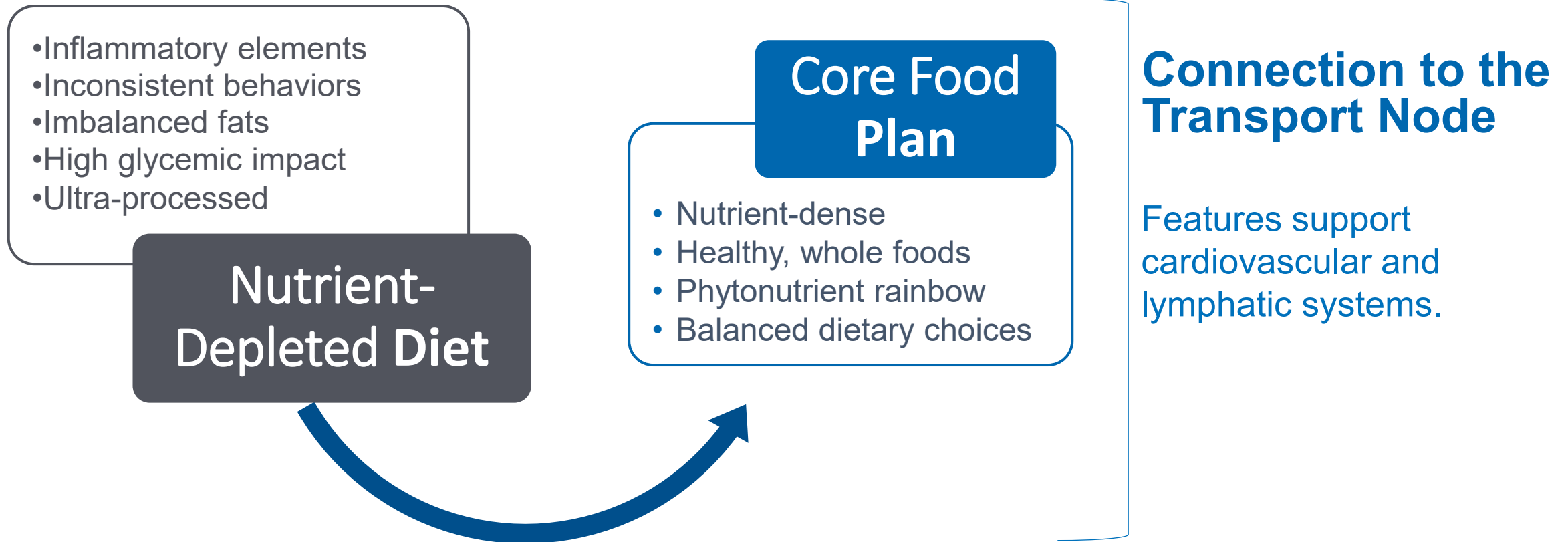
Dietary modifications that decrease insulin demand/release:

- Fiber (soluble and insoluble)
- “Good” (vs “bad”) fat
- “Good” (vs “bad”) carbohydrates
- “Good” protein

Management for Insulin Resistance

Recommended “Core” First Step for Most Patients

A More Nutrient-Dense Diet



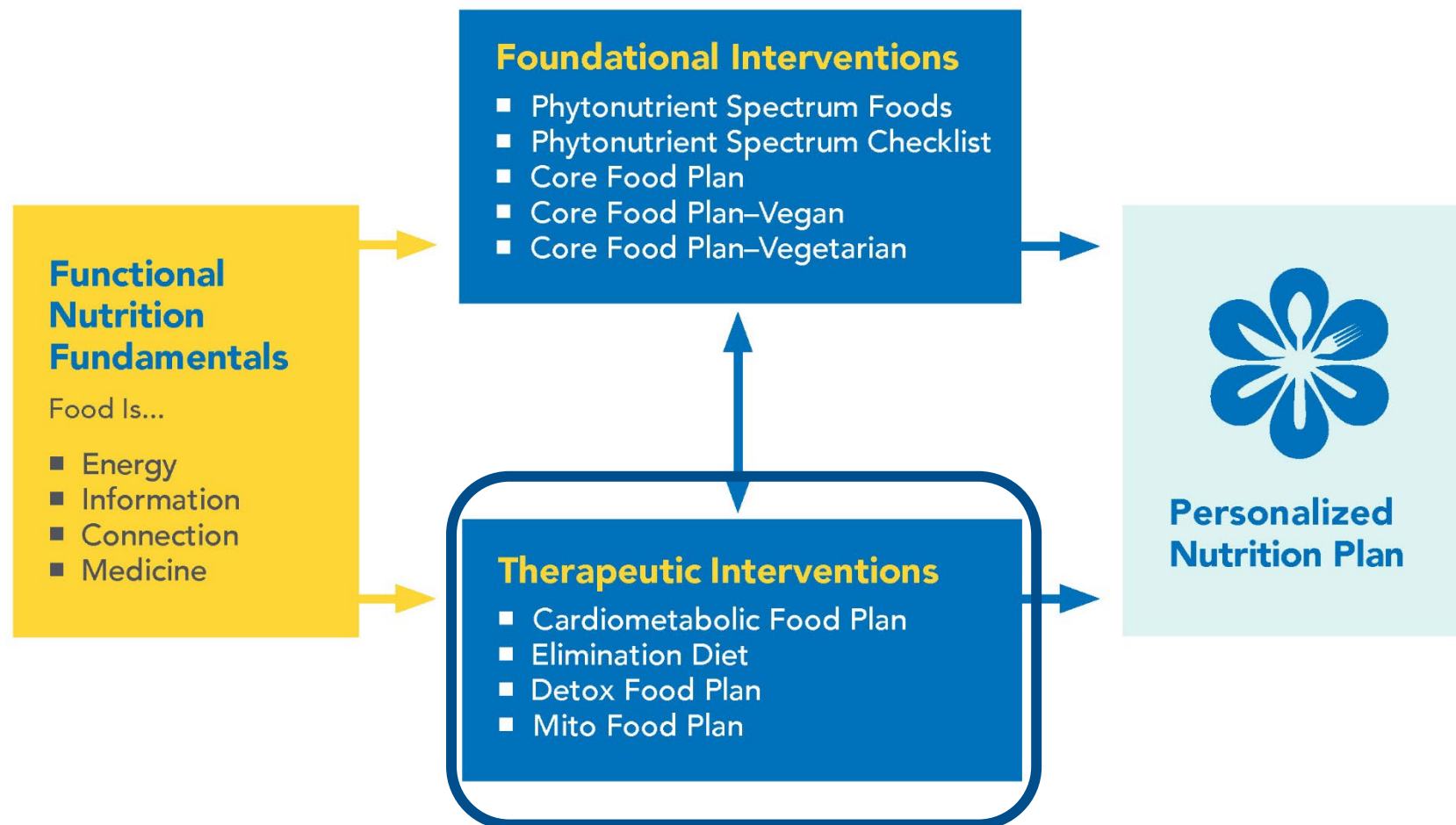
The Problem, the Solution, and the Goal

- Low-grade inflammation and insulin resistance play key roles in the pathophysiology of cardiovascular and metabolic diseases.
- A low-glycemic, anti-inflammatory, and phytonutrient-rich diet can help stabilize blood sugar levels, decrease chronic inflammation, and reduce insulin secretion.



As the patient's blood sugar stabilizes, both cardiovascular and metabolic disease are mitigated.

Nutrition Interventions



Long Description: nutrition therapeutic interventions

- Nutrition interventions
 - Use the functional nutrition fundamentals, foundational interventions, and therapeutic interventions to create a personalized nutrition plan.
- Functional Nutrition Fundamentals
 - Food is energy, information, connection, and medicine
- Foundational Interventions
 - Phytonutrient spectrum foods
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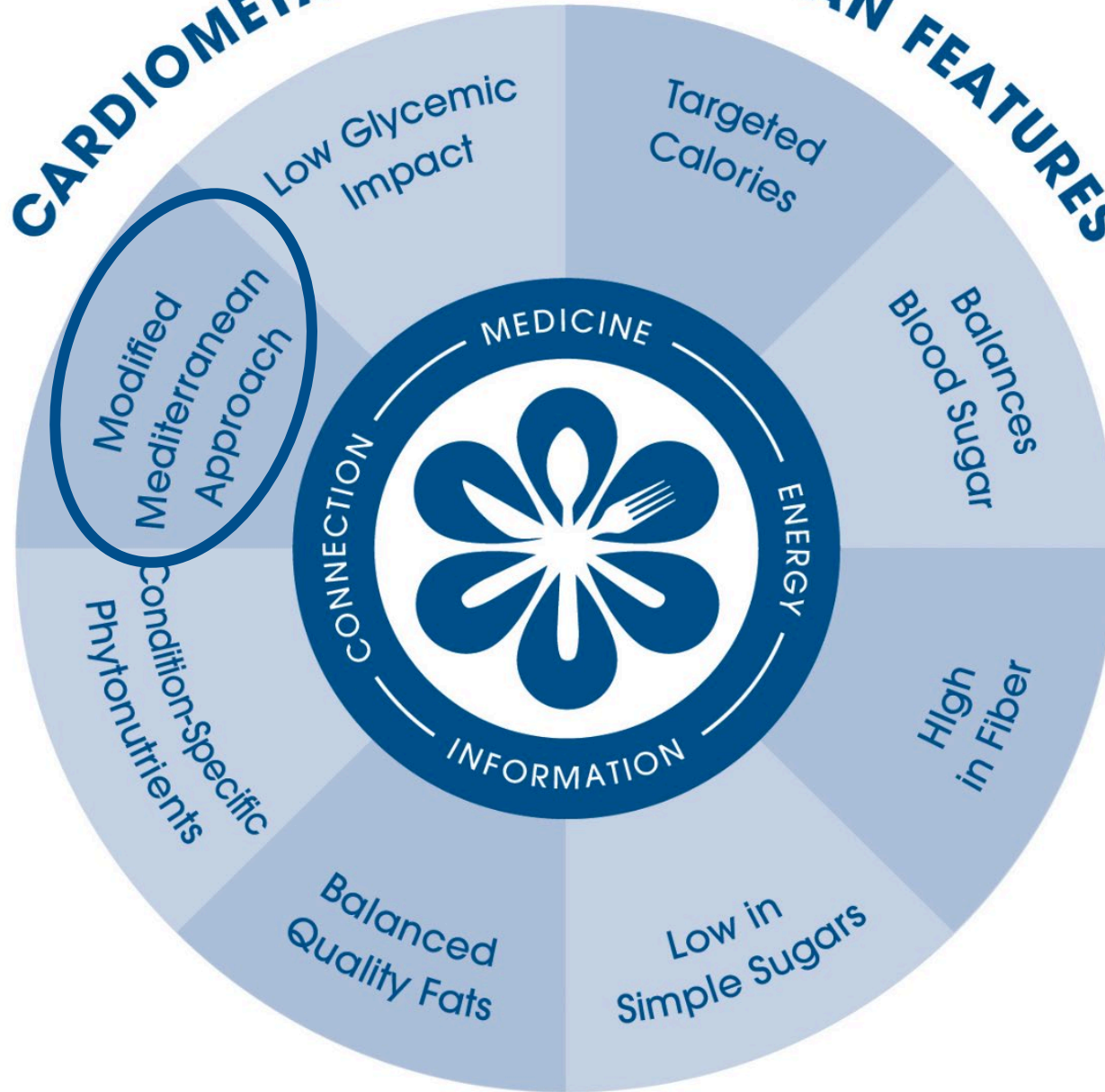
What Is the Cardiometabolic Food Plan?

A low glycemic, anti-inflammatory, phytonutrient-rich diet used to:

- Stabilize blood sugar
- Minimize insulin secretion
- Reduce chronic inflammation



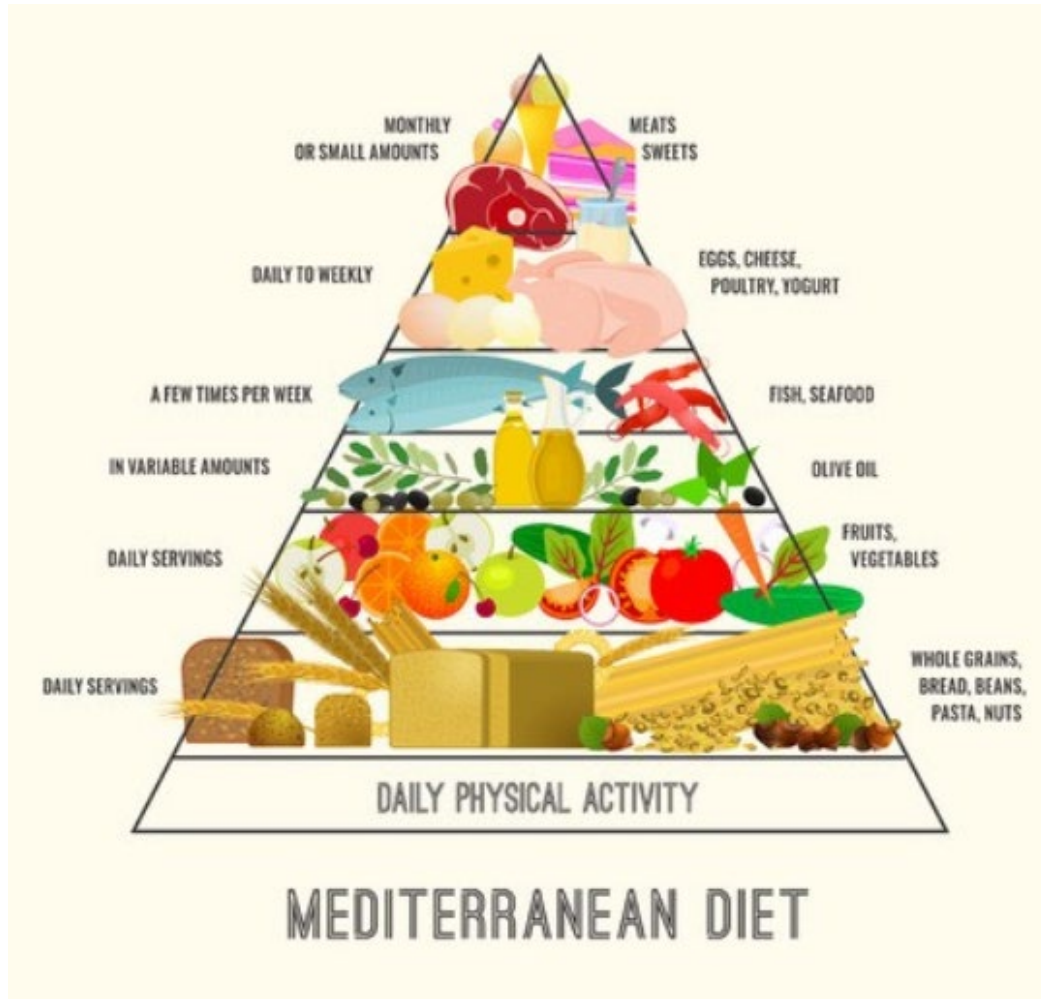
CARDIOMETABOLIC FOOD PLAN FEATURES



Long Description: Cardiometabolic Food Plan Features

- Cardiometabolic food plan features
 - Modified Mediterranean approach
 - Low glycemic impact
 - Targeted calories
 - Balances blood sugar
 - High in fiber
 - Low in simple sugars
 - Balanced quality fats
 - Condition-specific phytonutrients

Mediterranean-Style Diet

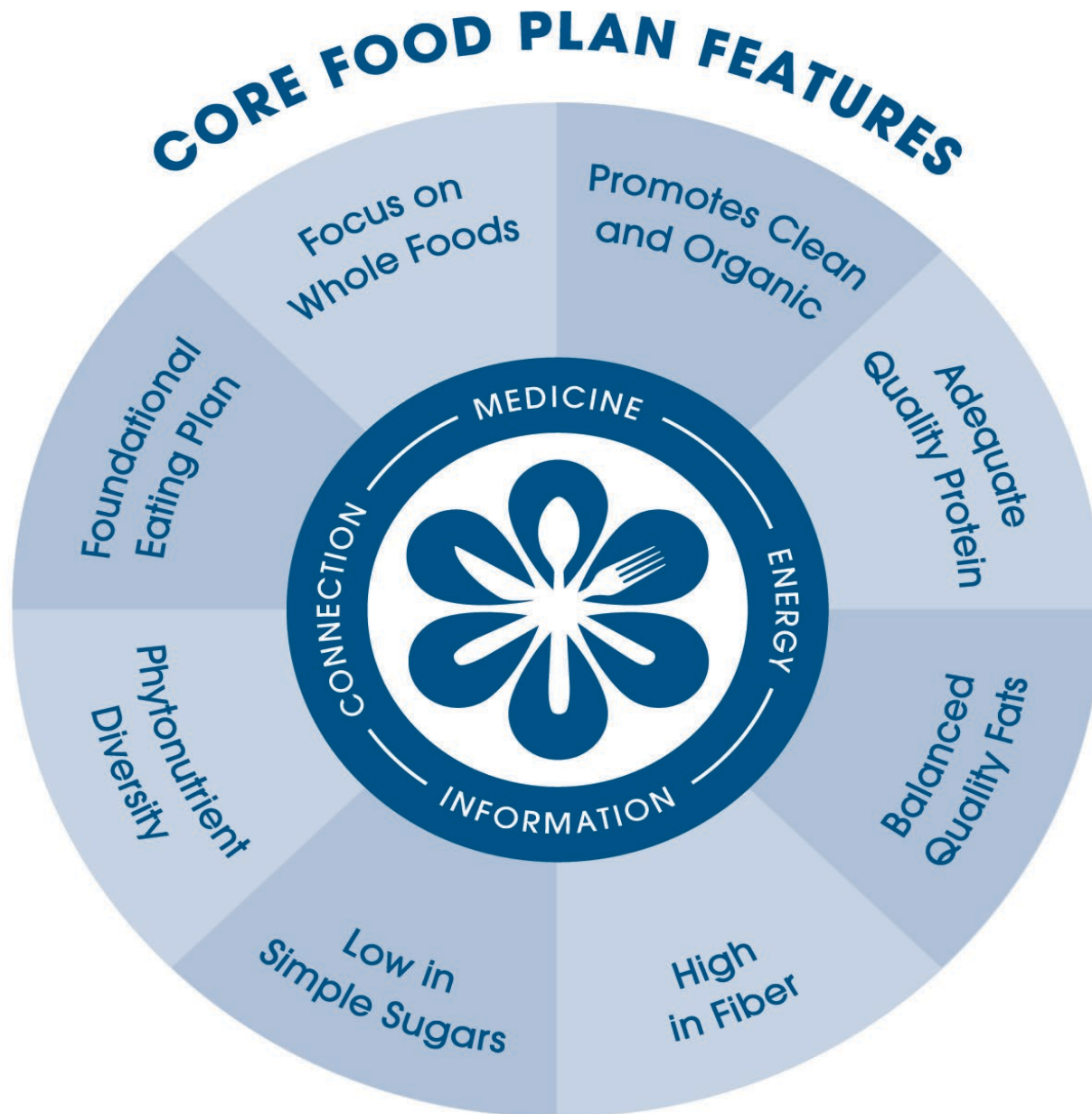


- High olive oil, vegetables, legumes, whole grains, fruit, nuts, spices, herbs
- Moderate fish and poultry
- Low red meat and dairy
- Low to moderate alcohol (e.g., red wine)

Core Food Plan & Cardiometabolic Food Plan

SIMILARITIES BETWEEN FOUNDATIONAL AND THERAPEUTIC FOOD PLANS

Core Food Plan	Cardiometabolic Food Plan
Focus on Whole Foods & Phytonutrient Diversity	Condition-Specific Phytonutrients, Low Glycemic Impact
Adequate Quality Protein	Balances Blood Sugar
Balanced Quality Fats	Balanced Quality Fats, Balances Blood Sugar
High Fiber	High in Fiber, Balances Blood Sugar
Low in Simple Sugars	Low in Simple Sugars, Low Glycemic Impact, Balances Blood Sugar
Targeted Calories (optional)	Targeted Calories
Dairy-Free, Gluten-Free (optional)	Dairy-Free, Gluten-Free (optional)



Accessible version in your toolkit.

Phytonutrient *Spectrum*



Comprehensive Guide

Long Description: Clinical toolkit items

- Core Food Plan Features
 - Focus on whole foods
 - Promotes clean and organic
 - Adequate quality protein
 - Balanced quality fats
 - High in fiber
 - Low in simple sugars
 - Phytonutrient diversity
 - Foundational eating plan
- Clinical Toolkit Item: Phytonutrient Spectrum
 - Accessible version available in course materials

Additional References

MEDITERRANEAN DIET AND DM

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NHS II:

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Clinical Focus

- Foundational gut health therapeutic interventions for cardiometabolic syndrome.



Foundational Therapeutic Approach

Transport Processes and Cardiometabolic Dysfunction



LIFESTYLE FIRST

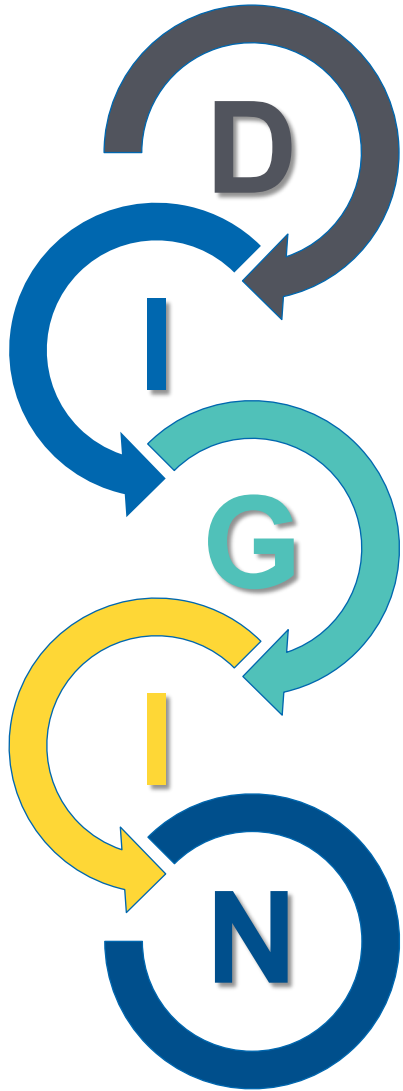


FOOD FIRST



GUT FIRST

The DIGIN Mnemonic of Gut Dysfunctions



Digestion/Absorption

Intestinal Permeability

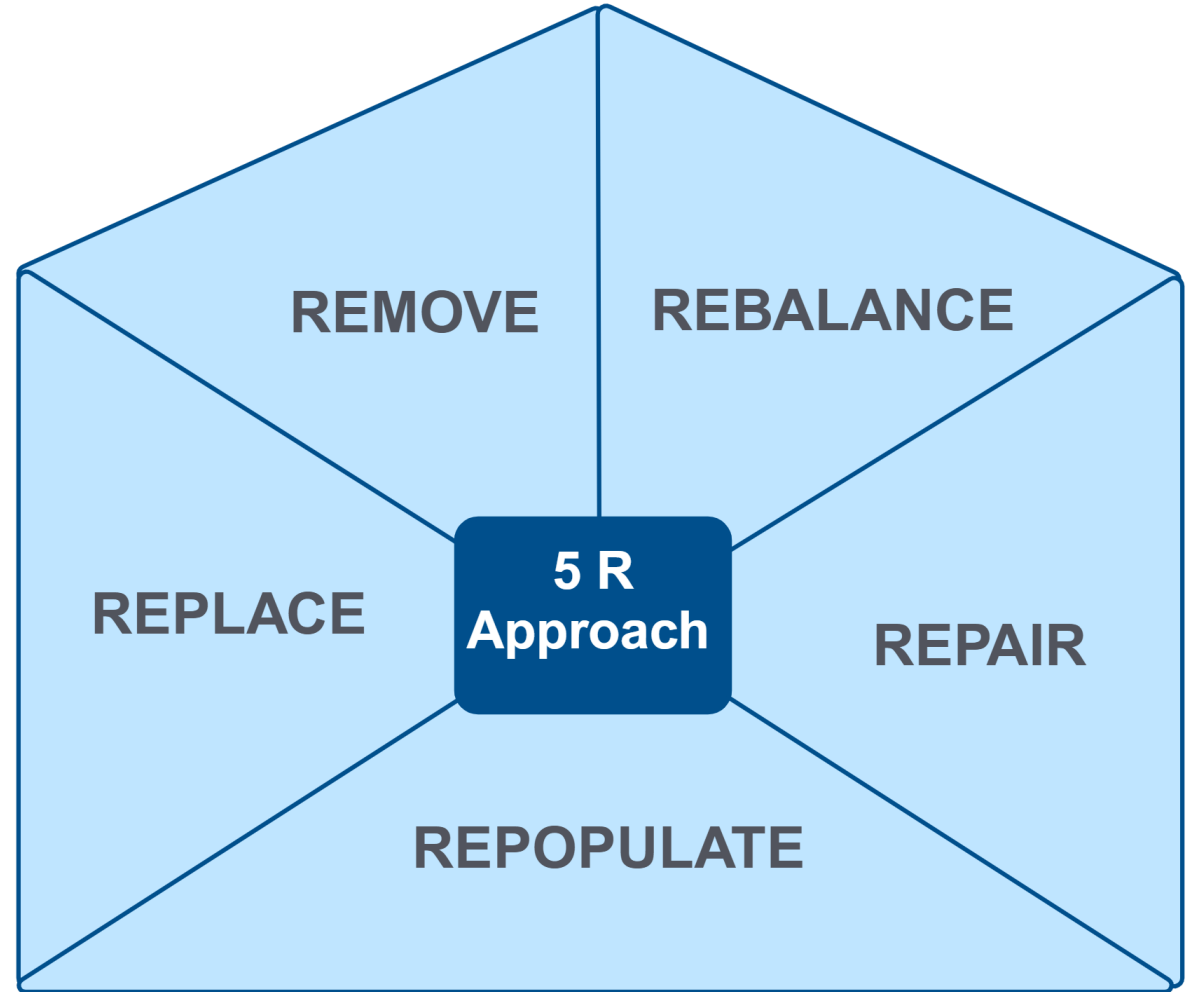
Gut Microbiota/Dysbiosis

Inflammation/Immune

Nervous System

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Therapeutic Interventions for Gut Dysfunction: The 5R Approach



Clinical Focus

- Foundational supplements and micronutrients for cardiometabolic syndrome.

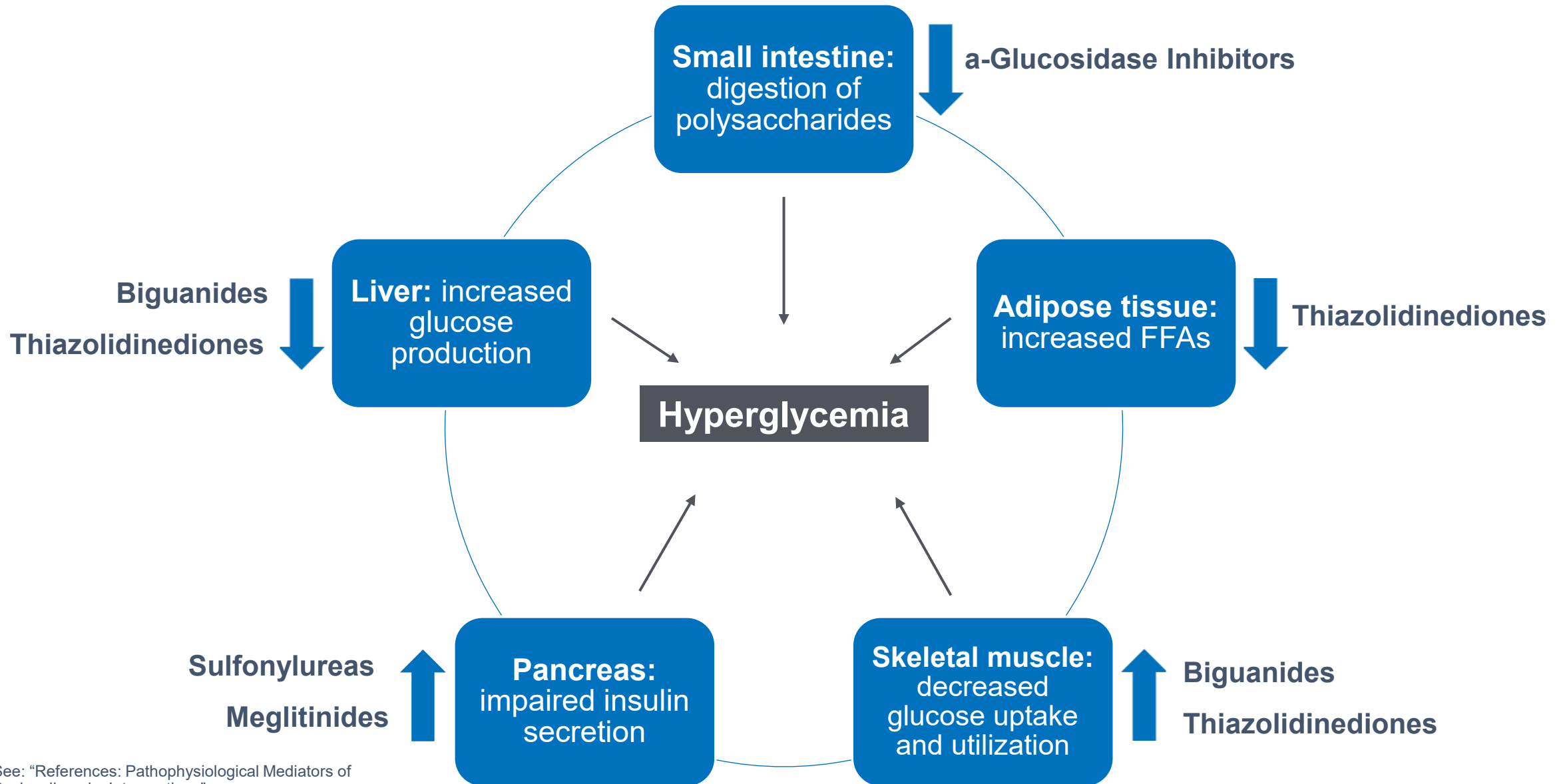


Foundational Therapeutic Interventions

- Referrals that address ATMs; address the matrix
- Lifestyle modification
- Dietary modifications
- Gut health
- **Foundational supplements – to increase cellular responsiveness to insulin (i.e., sensitivity)**

Pathophysiological Mediators of Dysinsulinemia

PHARMACEUTICAL MECHANISMS OF ACTION

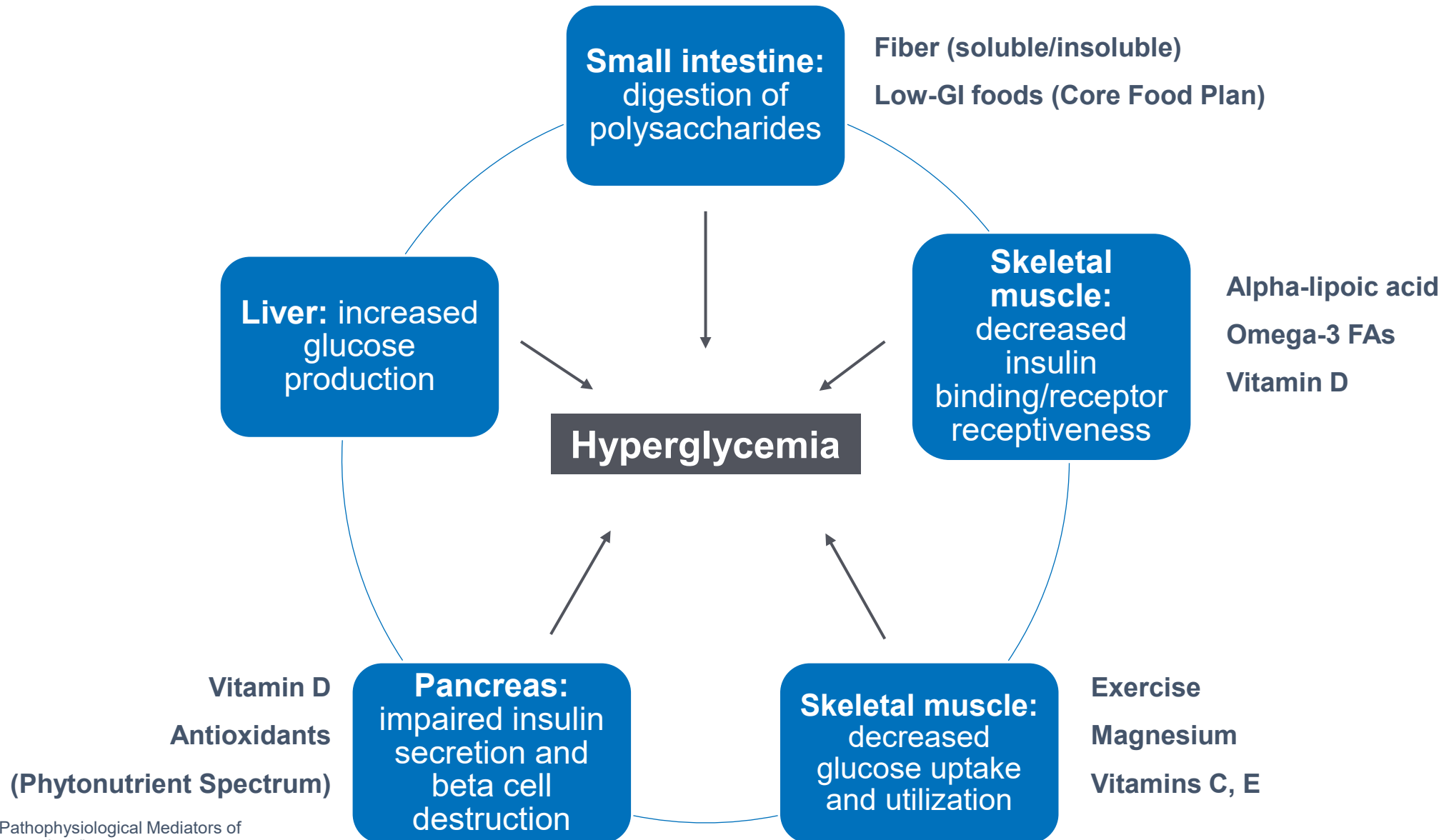


Long Description: Pathophysiological Mediators of Dysinsulinemia

- Alpha glucosidase inhibitors
 - slow digestion of polysaccharides
- Thiazolidinediones
 - Decrease free fatty acids
 - Decrease glucose production in the liver
 - Increase glucose uptake and utilization in skeletal muscle
- Biguanides
 - Decrease glucose production in the liver
 - Increase glucose uptake and utilization in skeletal muscle
- Sulfonylureas
 - Stimulate insulin secretion in the pancreas
- Meglitinides
 - Stimulate insulin secretion in the pancreas

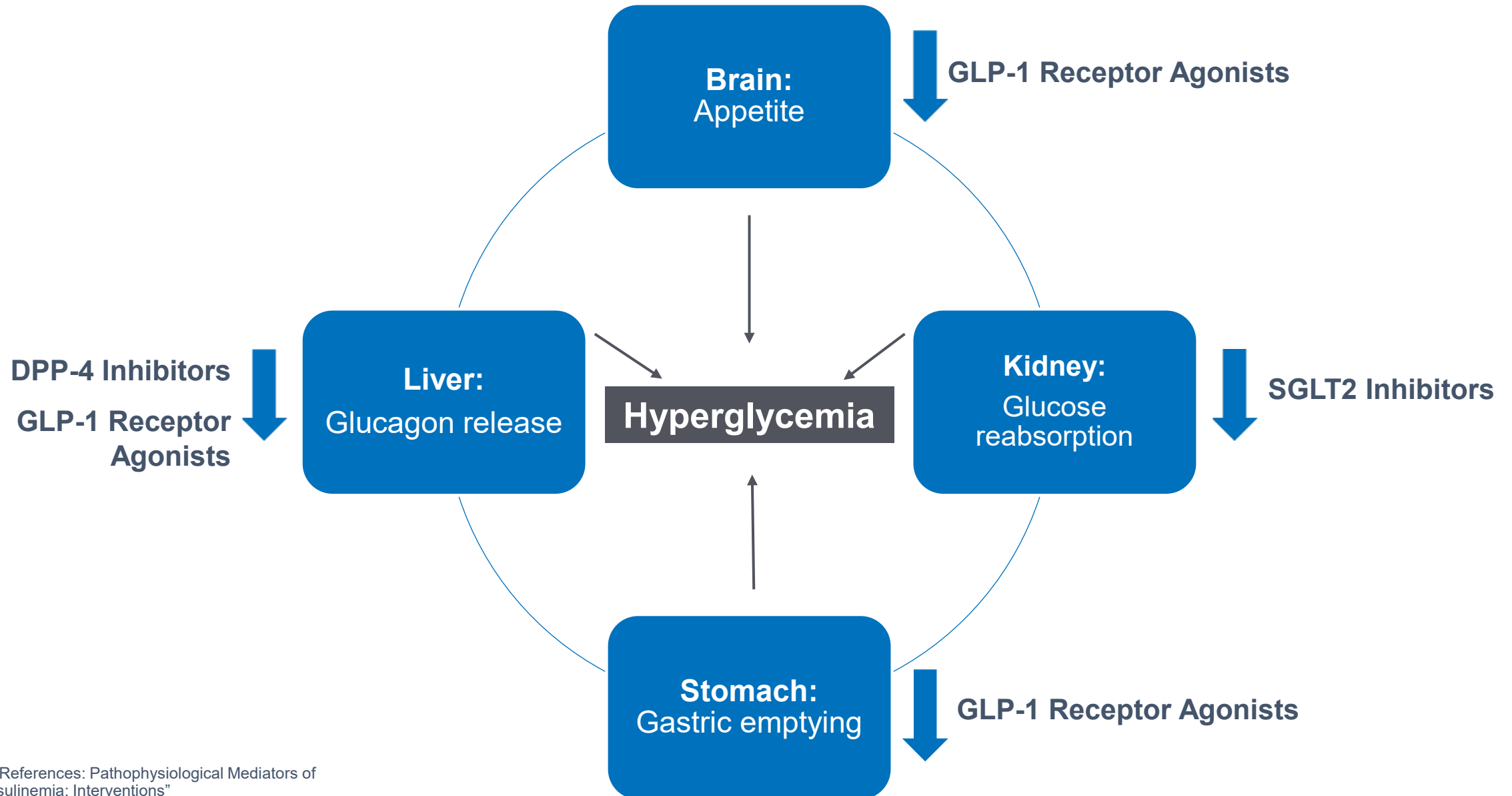
Pathophysiological Mediators of Dysinsulinemia

DIETARY, LIFESTYLE, AND NUTRACEUTICAL INTERVENTIONS



Pathophysiological Mediators of Dysinsulinemia

PHARMACEUTICAL MECHANISMS OF ACTION

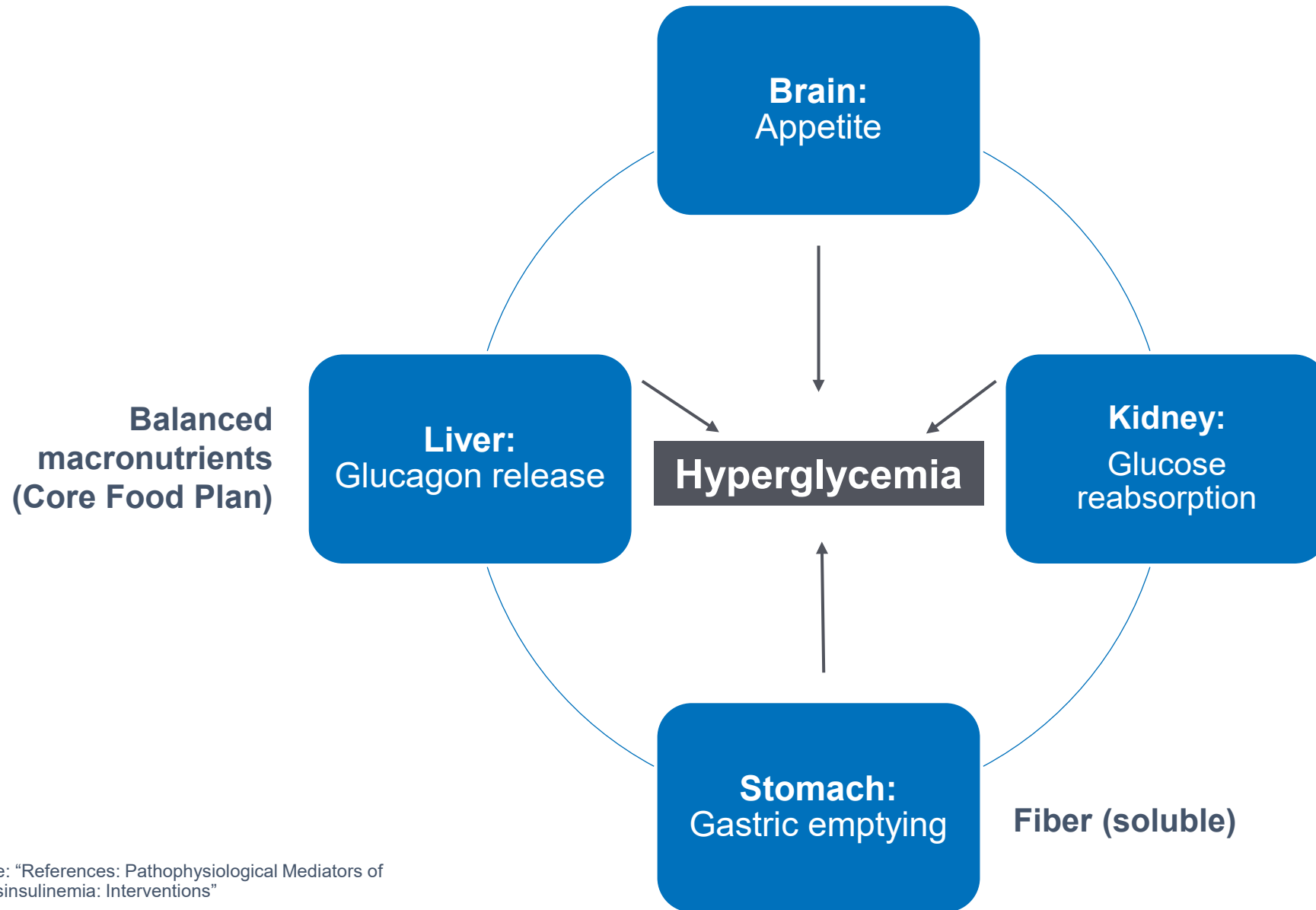


Long Description: pathophysiological mediators of dyslipidemia

- G L P 1 receptor agonists
 - Increase satiety signaling in the brain
 - Slows gastric emptying
 - Suppress glucagon
- S G L T 2 inhibitors
 - Decrease glucose reabsorption in the kidney
- D P P 4 inhibitors
 - Suppress glucagon

Pathophysiological Mediators of Dysinsulinemia

DIETARY, LIFESTYLE, AND NUTRACEUTICAL INTERVENTIONS



Long Description: pathophysiological mediators of dyslipidemia

- Balanced macronutrients, such as in the Core food plan
 - Suppress glucagon
- Soluble fiber
 - Slows gastric emptying

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PATHOPHYSIOLOGICAL MEDIATORS OF DYSINSULINEMIA: INTERVENTIONS

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Micronutrients to Increase Cellular Responsiveness to Insulin

Agents that modify insulin sensitivity at the cellular level:

- Alpha-lipoic acid
- CoQ10
- Vitamin D
- Vitamins C and E and other antioxidants
- Magnesium
- Omega-3 fatty acids

Management for Insulin Resistance

Foundational Micronutrient Recommendations

Micronutrient	Recommendation(s)	Notes
Omega-3s (DHA/EPA)	• 2 daily servings of fish OR • 1 g/day	Perform nutrition exam; food sources should be adequate.
Vitamin D	• 2,000-4,000 IUs/day	Test 25(OH)D; supplement as appropriate.
Magnesium	• See Food Sources of Magnesium for dietary recommendations • 200-400 mg/day, generally	Likely most effective if deficient, but accurate testing is cumbersome; supplement if signs and symptoms of deficiency/insufficiency are present.
Vitamin C	• Sweet peppers, citrus fruits, strawberries, broccoli, potatoes, Brussels sprouts	Perform nutrition exam; food sources should be adequate.
Zinc	• See Food Sources of Zinc for dietary recommendations	Perform nutrition exam; food sources should be adequate.

Foundational Micronutrient Recommendations

Micronutrient	Recommendation(s)	Notes
Selenium	See Food Sources of Selenium for dietary recommendations	Perform nutrition exam; food sources should be adequate.
Alpha-lipoic acid	Organ meats, beef, brewer's yeast, spinach, Brussels sprouts, rice bran, potatoes, broccoli, tomatoes	Food sources can be adequate.
CoQ10	Organ meats, beef, chicken, pork, mackerel, salmon, trout, sardines, almonds, sesame seeds, pistachios	Food sources can be adequate.

References

MICRONUTRIENTS TO INCREASE CELLULAR RESPONSIVENESS TO INSULIN

Dietary & Nutraceuticals

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IFM Toolkit Items



Glycemic Index and Glycemic Load

Maintaining healthy blood sugar levels is important for overall health. Diabetes, prediabetes, hypertension, heart disease, and other chronic conditions can be caused or worsened by poor blood sugar balance. The same impact on blood sugar and glycemic load (GL) are two common factors that affect blood sugar balance.

The Glycemic Index (GI) is a measure of how quickly a food raises blood sugar levels. The GI ranges from 0 to 100 based on how quickly a food's impact on blood sugar is absorbed. A score of 55 or less is considered low.

The Glycemic Load (GL) is calculated by multiplying the GI of a food by the number of grams of carbohydrates in a serving. A score of 10 or less is considered low.

Generally speaking, eating a diet with a low GI and GL helps to stabilize blood sugar levels, maintain a healthy weight, and reduce the risk of chronic diseases.

Low	
Glycemic Index	55 or less
Glycemic Load	10 or less

Tips for Blood Sugar Balance:

- While low GI foods are excellent for blood sugar balance, it's important to remember that high GI foods should be eaten in moderation and balanced with other nutrients on a daily basis to avoid sugar imbalances.

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Food Sources:

Essential Fatty Acids

Essential Fatty Acids (EFAs) cannot be created by your body. They are called "essential" because you must get them through your diet. There are two primary families of essential fatty acids: omega-6 and omega-3.

Linoleic acid (LA) and arachidonic acid (AA) are two common omega-6 fatty acids. LA is found in most plant oils, and AA is present in meat, poultry, and eggs. Alpha-linolenic acid (ALA), eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA) are the most common omega-3 fatty acids. Some plant foods contain ALA, while fatty fish are the best source of EPA and DHA.

Every cell in the body needs EFAs for proper structure and function. EFAs promote healthy nerve functioning and play a role in inflammation.

Food, Standard Serving Size	Amount of LA Omega-6s (in grams)
Walnuts, 1 ounce (14 halves)	10.8
Safflower oil, 1 Tbsp	10.1
Sunflower seeds (oil roasted), 1 ounce (1/4 cup)	9.7
Grapeseed oil, 1 Tbsp	9.5
Pine nuts, 1 ounce (1/4 cup)	9.4
Sunflower oil, 1 Tbsp	8.9
Hemp seeds (hulled), 3 Tbsp	8.3
Pecans (oil roasted), 1 ounce (19 halves)	6.4
Pumpkin seeds (dried), 1 ounce (1/4 cup)	5.9
Sesame oil, 1 Tbsp	5.6

Food, Standard Serving Size	Amount of Omega-3s (in grams)
Flaxseed oil, 1 Tbsp	7.3 (ALA)
Chia seeds, 1 ounce (2 Tbsp)	5.1 (ALA)
Hemp seeds (hulled), 3 Tbsp	2.6 (ALA)
Walnuts, 1 ounce (14 halves)	2.6 (ALA)
Flaxseed (whole), 1 Tbsp	2.4 (ALA)
Atlantic salmon (cooked), 3 ounces	1.8 (EPA and DHA)
Atlantic herring (cooked), 3 ounces	1.7 (EPA and DHA)
Sardines (canned), 3 ounces	1.2 (EPA and DHA)
Atlantic mackerel (cooked), 3 ounces	1.0 (EPA and DHA)

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2. U.S. Department of Agriculture, Agricultural Research Service, National Nutrient Database for Standard Reference, Legacy Version. <https://fdc.nal.usda.gov/>. Accessed April 2, 2024.
3. Oregon State University, Linus Pauling Institute, Micronutrient Information Center. Essential fatty acids. <https://lpi.oregonstate.edu/instruction/nutrition/essential-fatty-acids/introduction>. Reviewed May 2014. Accessed April 2, 2024.



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Food Sources:

Zinc

Zinc is an essential trace mineral; it must be absorbed through the diet and is required by the body in small amounts. Zinc is found in a variety of foods, including meat, poultry, fish, and shellfish. The bioavailability of zinc, or the amount that can be absorbed by the body, may be lower in grains and plant products, which bind to zinc and inhibit absorption.

Zinc plays an important role in several body functions. The body requires zinc to function. Zinc is involved in development, immune and neurological function, and maintaining the structure of cell membranes and oxidative damage.

Food, Standard Serving Size
Eastern oysters (breaded, fried), 3 ounces/6 medium
Eastern oysters (raw), 3 ounces/6 medium
Game meat, bison/deer (braised), 3 ounces
Beef chuck roast (braised), 3 ounces
Alaska king crab (cooked), 3 ounces
Ground beef (cooked), 3 ounces
Lamb shoulder (braised), 3 ounces
Pork shoulder (braised), 3 ounces
Breakfast cereal (fortified), 1/4 cup
Lobster (cooked), 3 ounces
Baked beans (canned), 1/2 cup
Pork loin (braised), 3 ounces
Chicken, dark meat (roasted), 3 ounces
Pumpkin seeds (dried), 1 ounce/~1/4 cup
Cashews (dry roasted), 1 ounce/~17 nuts

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3. Oregon State University, Linus Pauling Institute, Micronutrient Information Center. Zinc. <https://lpi.oregonstate.edu/instruction/nutrition/zinc>. Updated June 11, 2015. Accessed March 23, 2020.



Food Sources:

Selenium

Selenium is an essential trace mineral important for thyroid health, DNA production, reproductive health, immune support, and antioxidant defense. The nutrient is found in both animal and plant-based foods.

Selenium is present in seafood, meats, eggs, some dairy products, grains, nuts, seeds, and other plant foods. Plant sources of the mineral may have variable selenium content, as their uptake of this nutrient depends on the selenium concentration of the soil where they are grown.

Food, Standard Serving Size*	Average Selenium Content (in micrograms)
Brazil nuts, 1 ounce (5-8 nuts)	544
Tuna (yellowfin), 3 ounces	92
Oysters (Pacific), 3 ounces	66
Cleams, 3 ounces	54
Halibut, 3 ounces	47
Sardines (canned), 3 ounces	45
Shrimp, 3 ounces	42
Salmon (wild-caught), 3 ounces	31
Cod (Atlantic), 3 ounces	28
Beef chuck roast, 3 ounces	23
Cottage cheese, 1 cup	20
Sunflower seed butter, 1 tablespoon	17
Egg, whole, hard-boiled	15
Oatmeal, cooked with water, 1 cup	13

*Serving size and nutrient amounts for all meat and fish are for the cooked foods.

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1. Oregon State University, Linus Pauling Institute, Micronutrient Information Center. Selenium. <https://lpi.oregonstate.edu/instruction/nutrition/selenium>. Updated June 2015. Accessed March 23, 2020.
2. U.S. Department of Health and Human Services, National Institutes of Health, Office of Dietary Supplements. Selenium. <https://ods.od.nih.gov/factsheets/SeleniumHealthProfessional/>. Updated March 11, 2020. Accessed March 23, 2020.
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Version 4, updated 2024

Accessible version of these resources
in your toolkit.

Long Description: IFM Toolkit items

- IFM Clinical Toolkit Items: Accessible versions available in course materials
 - Glycemic index and glycemic load
 - Food sources: essential fatty acids
 - Food sources: zinc
 - Food sources: selenium

Clinical Focus

- Conclusion and Key Takeaways
- Clinical Flow Path for Cardiometabolic Syndrome



Cardiometabolic Dysfunction: Clinical Takeaways



Impairment of glucose transport with resultant insulin resistance and excess is the primary physiological dysfunction.



This dysfunction is driven largely by mediators of adverse dietary and lifestyle factors.



Excess insulin acting on tissues that remain insulin sensitive mediates physiological dysfunction in many bodily systems.

Cardiometabolic Dysfunction: Clinical Takeaways



Addressing each patient's diet and lifestyle factors is essential for favorable therapeutic outcomes.

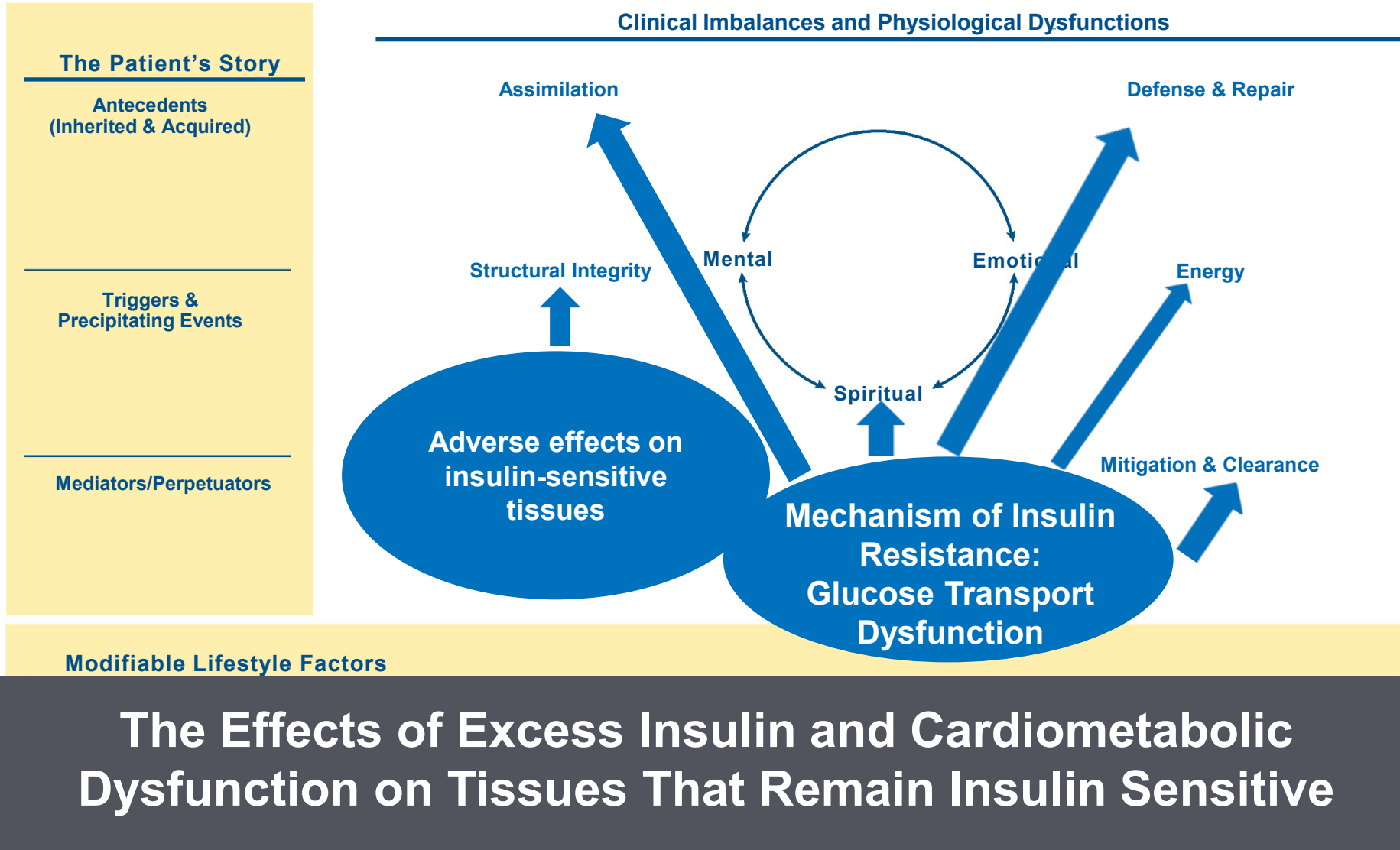


Initiate referrals to collaborative care team colleagues as needed to assist with these interventions.



Track the ABCDs (anthropometrics, biomarkers, clinical indicators, diet) and the patient's changing priorities.

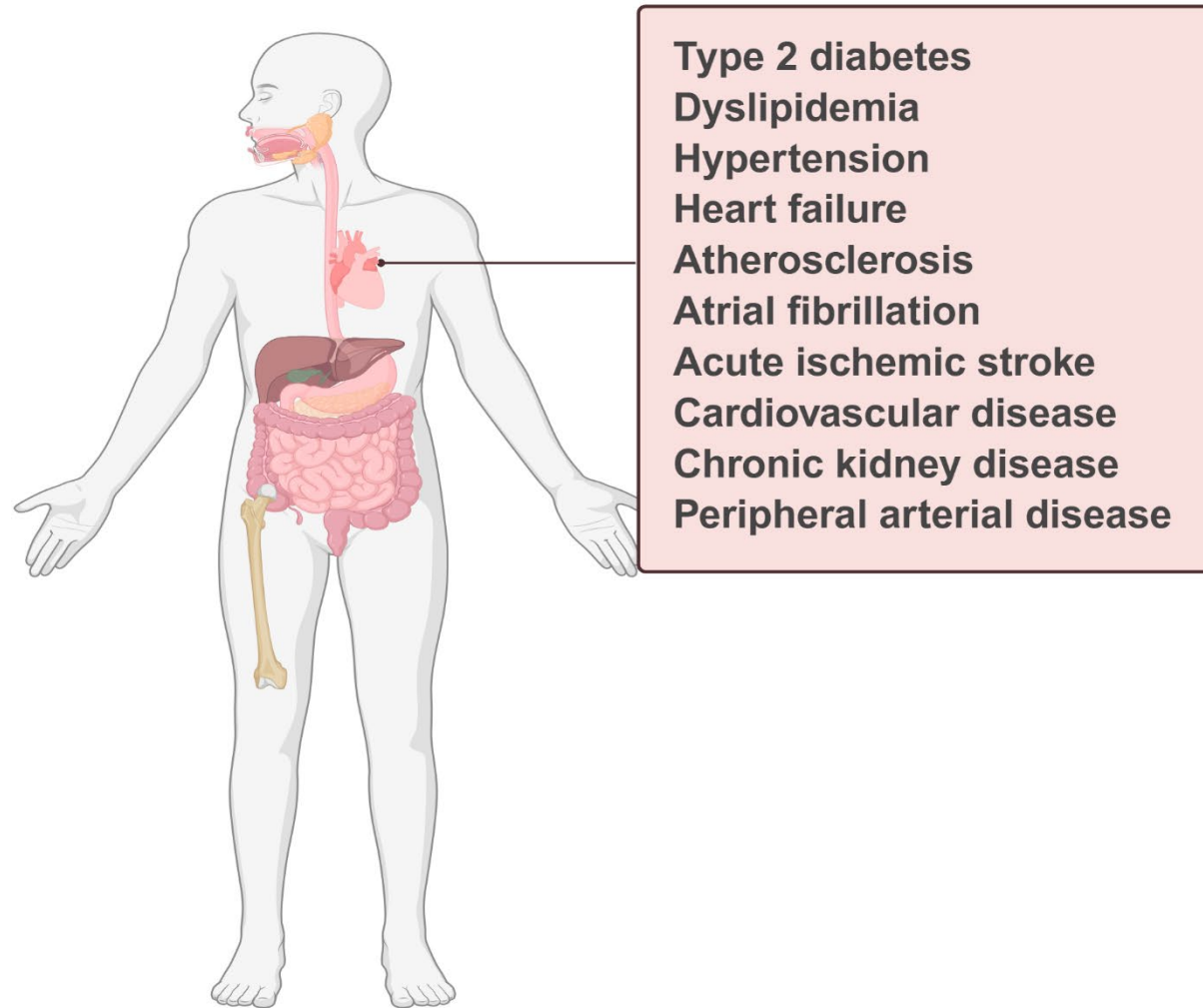
Functional Medicine Matrix



Long Description: **The Effects of Excess Insulin and Cardiometabolic Dysfunction on Tissues That Remain Insulin Sensitive**

Insulin resistance impacts the assimilation, defense and repair, energy, mitigation and clearance, mental, emotional, and spiritual nodes of the matrix. Adverse effects on insulin-sensitive tissues lives in the communication node and impacts the structural integrity node.

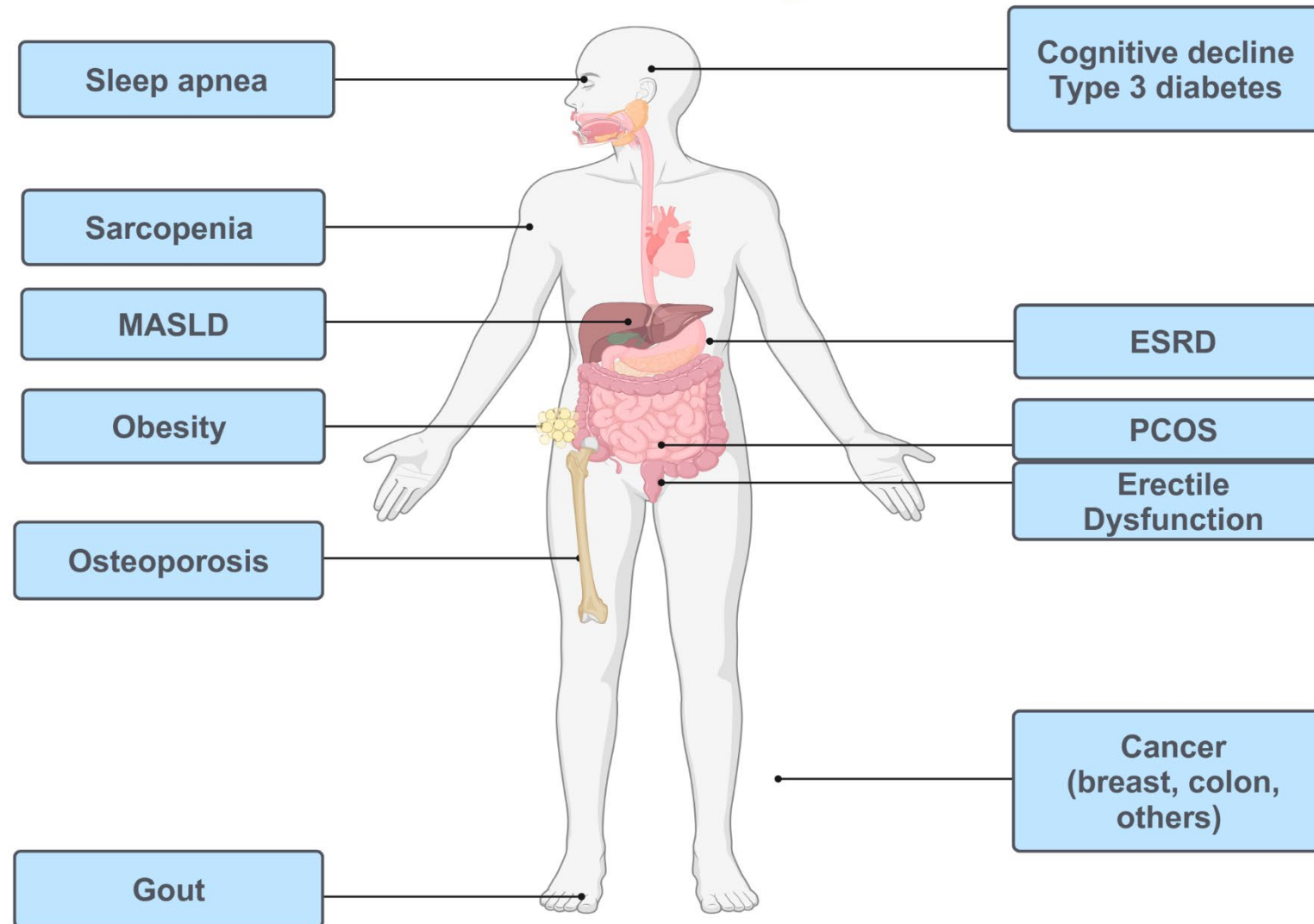
The “Cardio” in Cardiometabolic Syndrome: Chronic Conditions Linked to the Syndrome



Long Description: Chronic conditions linked to cardiometabolic syndrome

- Type 2 diabetes
- Dyslipidemia
- Hypertension
- Heart failure
- Atherosclerosis
- Atrial fibrillation
- Acute ischemic stroke
- Cardiovascular disease
- Chronic kidney disease
- Peripheral arterial disease

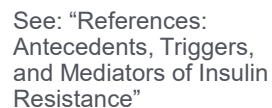
Beyond the “Cardio” in Cardiometabolic Syndrome: Chronic Conditions Linked to the Syndrome



Long Description: Beyond the "cardio" in cardiometabolic syndrome: chronic conditions linked to the syndrome

Chronic conditions linked to cardiometabolic syndrome beyond the cardiovascular system:

- Sleep apnea
- Cognitive decline in Type 3 diabetes
- Sarcopenia
- Metabolic dysfunction associated steatotic liver disease
- End-stage renal disease
- Obesity
- P C O S
- Erectile dysfunction
- Osteoporosis
- Gout
- Cancer



Long Description: ATMs of cardiometabolic syndrome

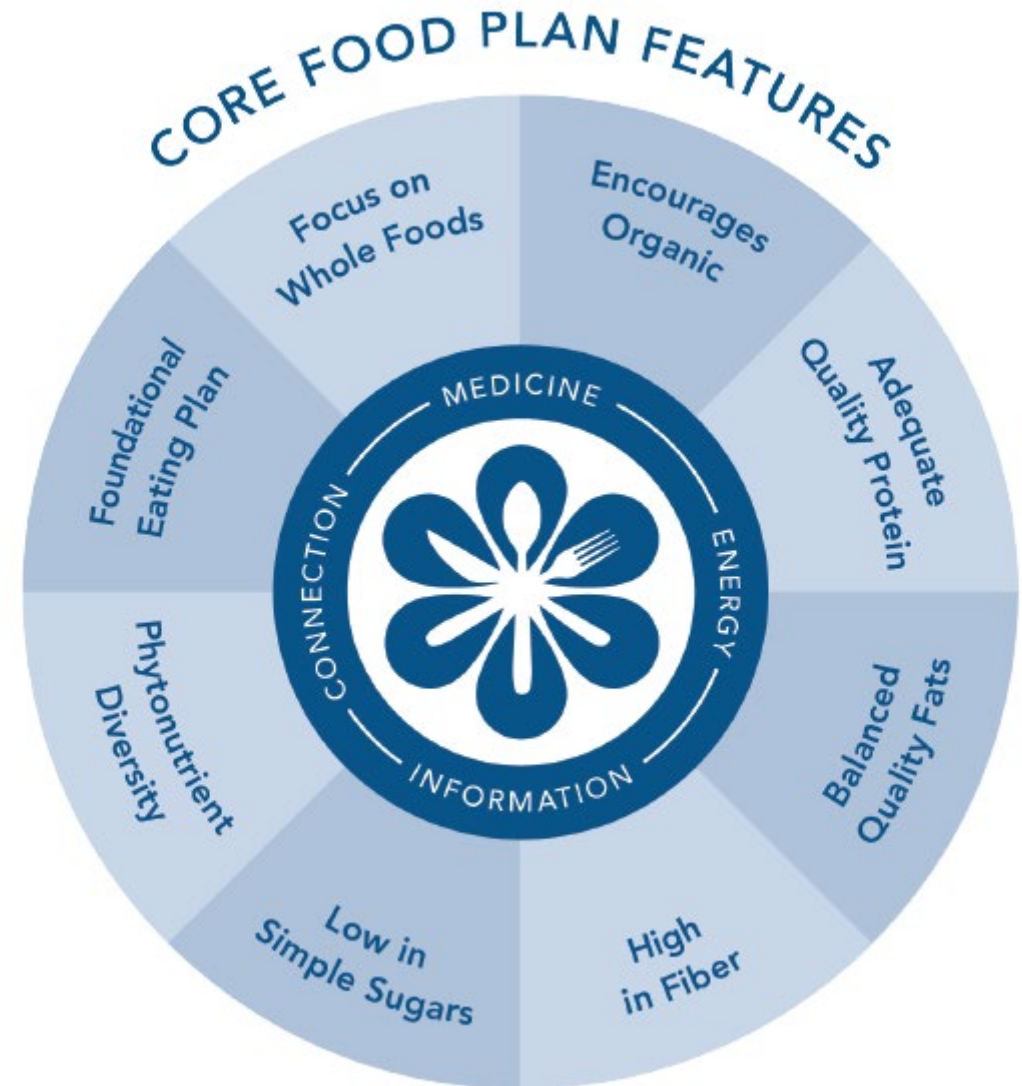
- Antecedents
 - Genetic predisposition / S N P's
 - A C E's
 - S D O H
- Triggers and precipitating events
 - Gestational diabetes
 - P O P's
 - Mitotoxins
- Mediators/perpetuators
 - Inflammation
 - S D O H
 - Dysbiosis
 - Toxins
 - V A T / Obesity
 - Smoking
 - Sarcopenia
- Modifiable Lifestyle Factors
 - Sleep
 - Suboptimal sleep
 - Physical Activity
 - Sedentary behaviors
 - Nutrition
 - High glycemic load
 - Nutrient insufficiencies
 - Stressors
 - Chronic stress
 - Relationships
 - Loneliness/social isolation

Overall Lifestyle Changes



FOUNDATIONAL FOOD INTERVENTIONS

Phytonutrient Spectrum



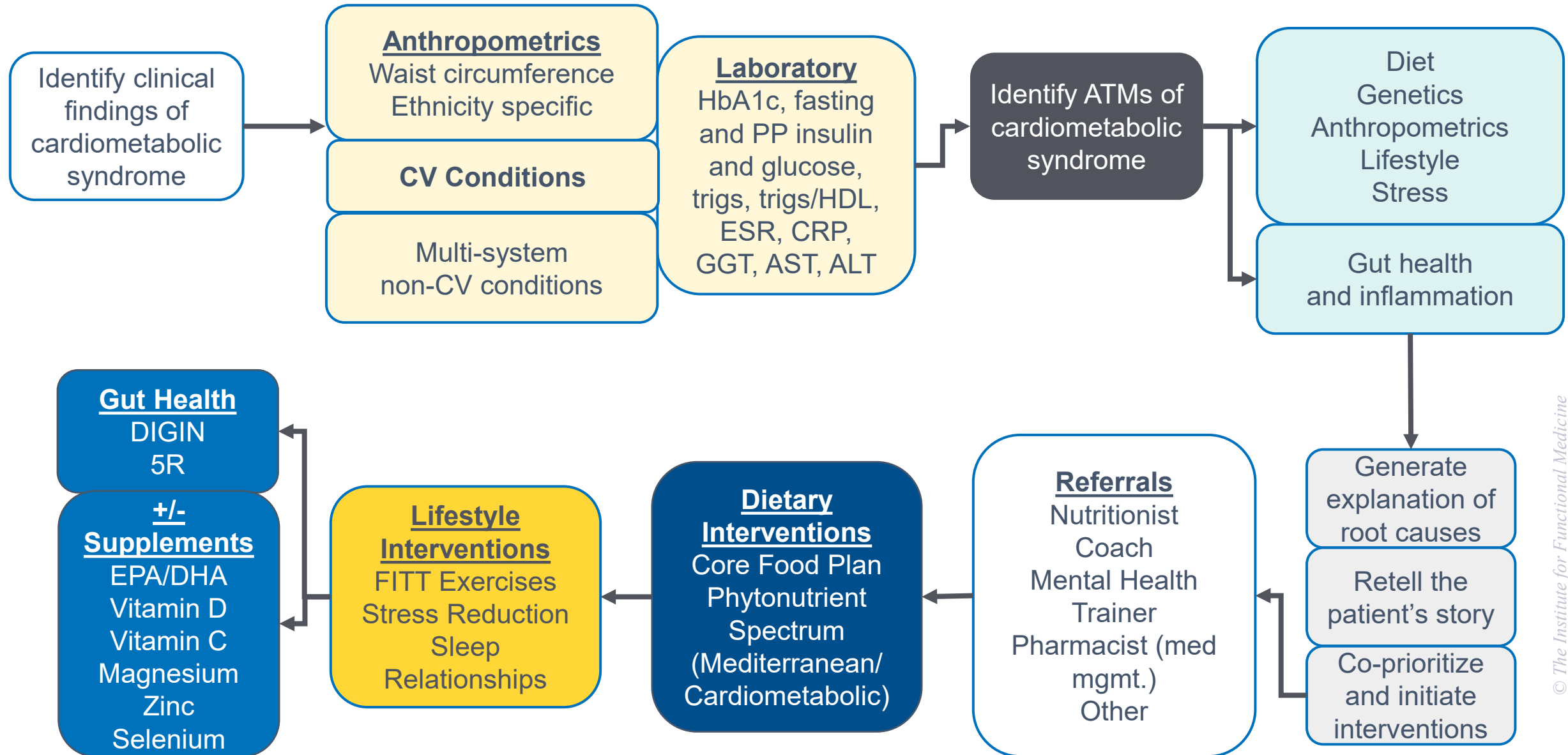
Foundational Micronutrient Recommendations

Micronutrient	Recommendation(s)	Notes
Omega-3s (DHA/EPA)	• 2 daily servings of fish OR • 1 g/day	Perform nutrition exam; food sources should be adequate.
Vitamin D	• 2,000-4,000 IUs/day	Test 25(OH)D; supplement as appropriate.
Magnesium	• See Food Sources of Magnesium for dietary recommendations • 200-400 mg/day, generally	Likely most effective if deficient, but accurate testing is cumbersome; supplement if signs and symptoms of deficiency/insufficiency are present.
Vitamin C	• Sweet peppers, citrus fruits, strawberries, broccoli, potatoes, Brussels sprouts	Perform nutrition exam; food sources should be adequate.
Zinc	• See Food Sources of Zinc for dietary recommendations	Perform nutrition exam; food sources should be adequate.

Foundational Micronutrient Recommendations

Micronutrient	Recommendation(s)	Notes
Selenium	See Food Sources of Selenium for dietary recommendations	Perform nutrition exam; food sources should be adequate.
Alpha-lipoic acid	Organ meats, beef, brewer's yeast, spinach, Brussels sprouts, rice bran, potatoes, broccoli, tomatoes	Food sources can be adequate.
CoQ10	Organ meats, beef, chicken, pork, mackerel, salmon, trout, sardines, almonds, sesame seeds, pistachios	Food sources can be adequate.

Flow Path: Cardiometabolic Syndrome



Long Description: Flow Path: Cardiometabolic Syndrome

- Identify clinical indicators of cardiometabolic syndrome
 - Anthropometrics: waist circumference, ethnicity specific
 - Cardiovascular conditions
 - Multi-system and non-cardiovascular conditions
 - Laboratory values: H B A 1 C, fasting and postprandial insulin and glucose, triglycerides, triglyceride H D L ratio, E S R, C R P, G G T, A S T, A L T
- Identify A T Ms of cardiometabolic syndrome
 - Diet, genetics, anthropometrics, lifestyle, stress
 - Gut health
- Generate explanation of root causes
- Retell the patient's story
- Co-prioritize and initiate interventions
- Referrals: Nutritionist, coach, mental health, trainer, pharmacist / medication management, other
- Dietary Interventions
 - Core food plan, Phytonutrient spectrum, Mediterranean, Cardiometabolic
- Lifestyle interventions
 - F I T T exercises
 - Stress reduction
 - Sleep
 - Relationships
- Gut Health
 - D I G I N and 5 R
- Add or remove supplements:
 - E P A and D H A
 - Vitamin D
 - Vitamin C
 - Magnesium
 - Zinc
 - Selenium

Transport Processes and Cardiometabolic Dysfunction

The Functional Medicine Matrix

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE